

Lesson 01: Lesson End Project

Migrating Database Using AWS DMS

Project agenda: To perform a structured migration of data from a MySQL Community Edition RDS instance to an Amazon Aurora MySQL cluster using AWS Database Migration Service (DMS). This process involves provisioning and configuring the source RDS instance and the target Aurora cluster, establishing secure connectivity between them, and setting up AWS DMS components, including the replication instance, source and target endpoints, and migration tasks.

Description: You are tasked with performing a database migration from a MySQL Community Edition RDS instance to an Amazon Aurora MySQL instance using AWS Database Migration Service (DMS). This project involves setting up the source and target endpoints, configuring replication tasks, and ensuring secure connectivity between the databases. The goal is to minimize downtime during migration, validate schema and data consistency, and simulate real-world replication scenarios. By using AWS DMS, this setup provides a hands-on demonstration of cloud-native database migration practices, helping you understand how to move workloads to Aurora while maintaining application availability.

Tools required: AWS RDS, AWS DMS

Prerequisites: None

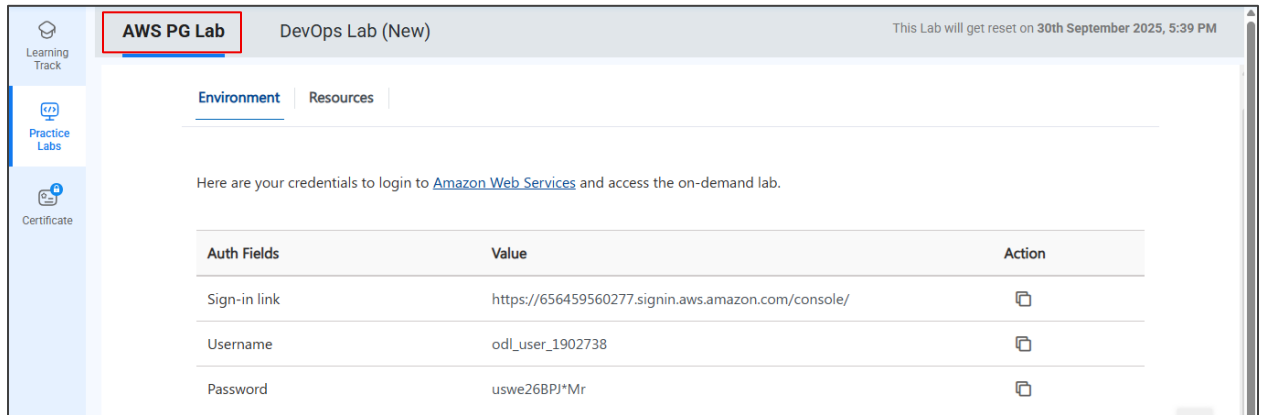
Expected deliverables: A fully configured source MySQL RDS instance and target Aurora MySQL cluster, a detailed AWS DMS setup guide covering prerequisites, IAM roles, endpoints, replication instances, and tasks, a successfully executed replication task performing full load and ongoing data replication, a validation report confirming schema and data consistency between source and target, documented observations on migration performance and downtime considerations, and a final migration report summarizing the entire process with identified issues, applied fixes, and recommendations for production-grade database migrations.

Steps to be followed:

1. Create MySQL and Aurora DB instances in the RDS service
2. Create a replication instance in AWS DMS
3. Implement database migration using AWS DMS

Step 1: Create MySQL and Aurora DB instances in the RDS service

1.1 Launch AWS PG Lab

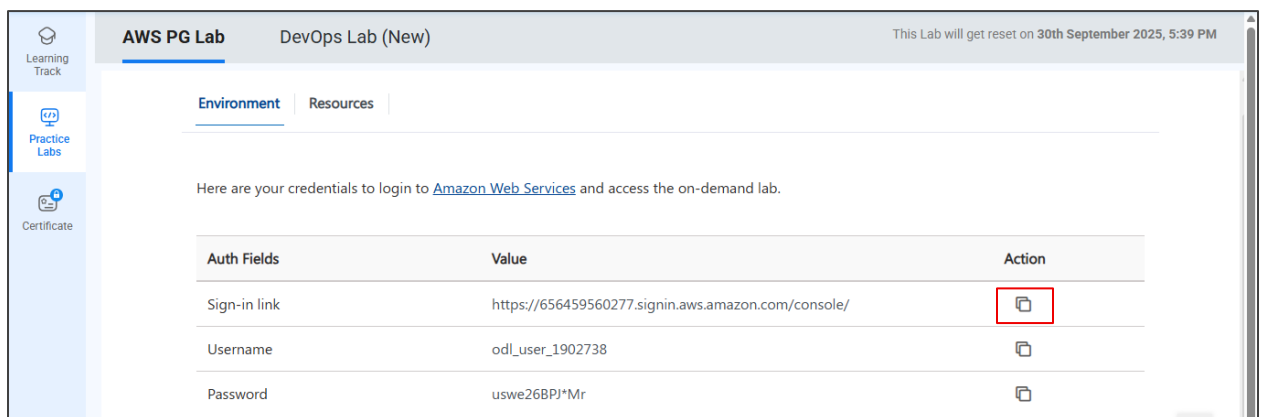


The screenshot shows the AWS PG Lab interface. The left sidebar contains 'Learning Track', 'Practice Labs', and 'Certificate'. The main header shows 'AWS PG Lab' (highlighted with a red box) and 'DevOps Lab (New)'. A message at the top right states: 'This Lab will get reset on 30th September 2025, 5:39 PM'. The 'Environment' tab is active, displaying a table of credentials.

Here are your credentials to login to [Amazon Web Services](#) and access the on-demand lab.

Auth Fields	Value	Action
Sign-in link	https://656459560277.signin.aws.amazon.com/console/	
Username	odl_user_1902738	
Password	uswe26BPJ*Mr	

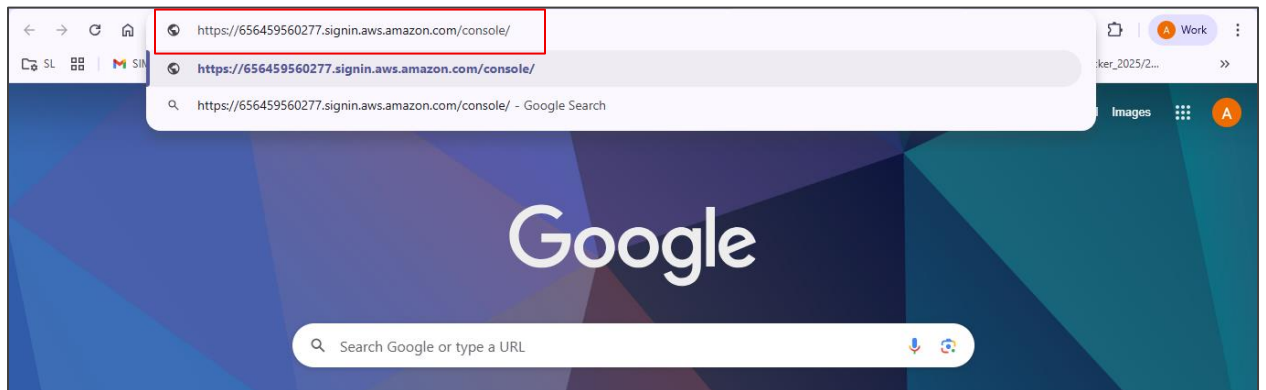
1.2 Click the Copy sign-in link button



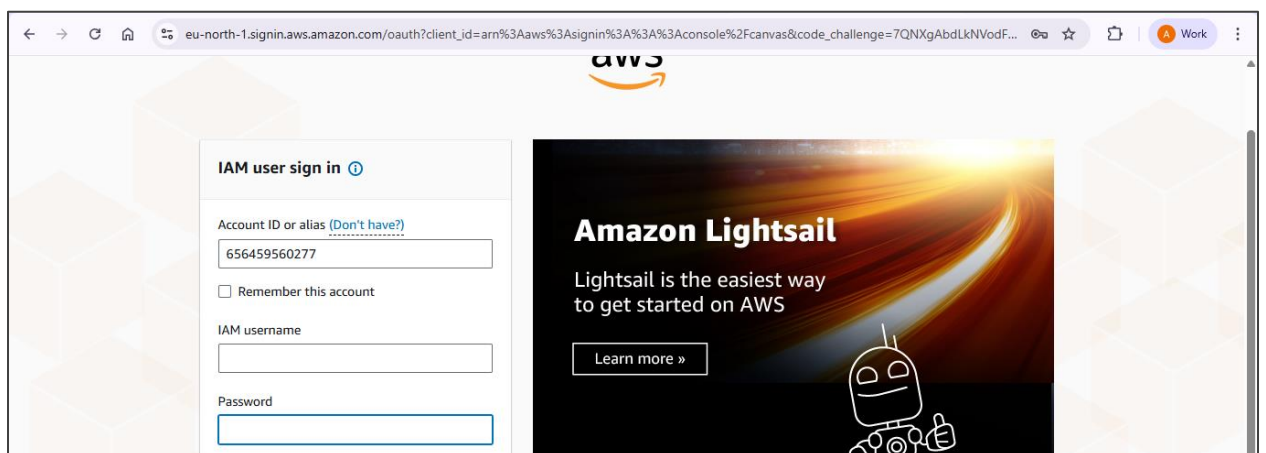
This screenshot is identical to the previous one, but with a red box highlighting the 'Copy' icon in the 'Action' column for the 'Sign-in link' row.

Auth Fields	Value	Action
Sign-in link	https://656459560277.signin.aws.amazon.com/console/	
Username	odl_user_1902738	
Password	uswe26BPJ*Mr	

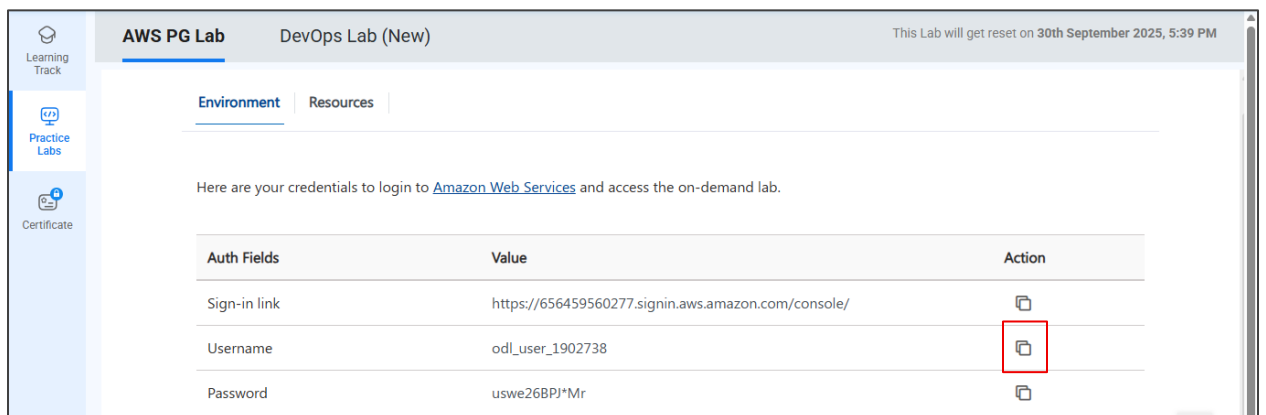
1.3 Paste the copied link into the browser's address bar and press Enter to open the AWS Console login page



The following page opens:



1.4 Click the copy icons to copy the AWS console Username and Password



Learning Track

Practice Labs

Certificate

AWS PG Lab

DevOps Lab (New)

This Lab will get reset on 30th September 2025, 5:39 PM

Environment

Resources

Here are your credentials to login to [Amazon Web Services](#) and access the on-demand lab.

Auth Fields	Value	Action
Sign-in link	https://656459560277.signin.aws.amazon.com/console/	
Username	odL_user_1902738	
Password	uswe26BPJ*Mr	

1.5 Paste the copied IAM username and password, and then click the **Sign in** button to access the AWS Console

← → ↺ 🏠

eu-north-1.signin.aws.amazon.com/oauth?client_id=arn%3Aaws%3Asignin%3A%3A%3Aconsole%2Fcanvas&code_challenge=7QNXgAbdLkNVodF...

🔍 ⭐ 📁 Work

IAM user sign in ⓘ

Account ID or alias [\(Don't have?\)](#)

656459560277

☐ Remember this account

IAM username

odL_user_1902738

Password

☐ Show Password [Having trouble?](#)

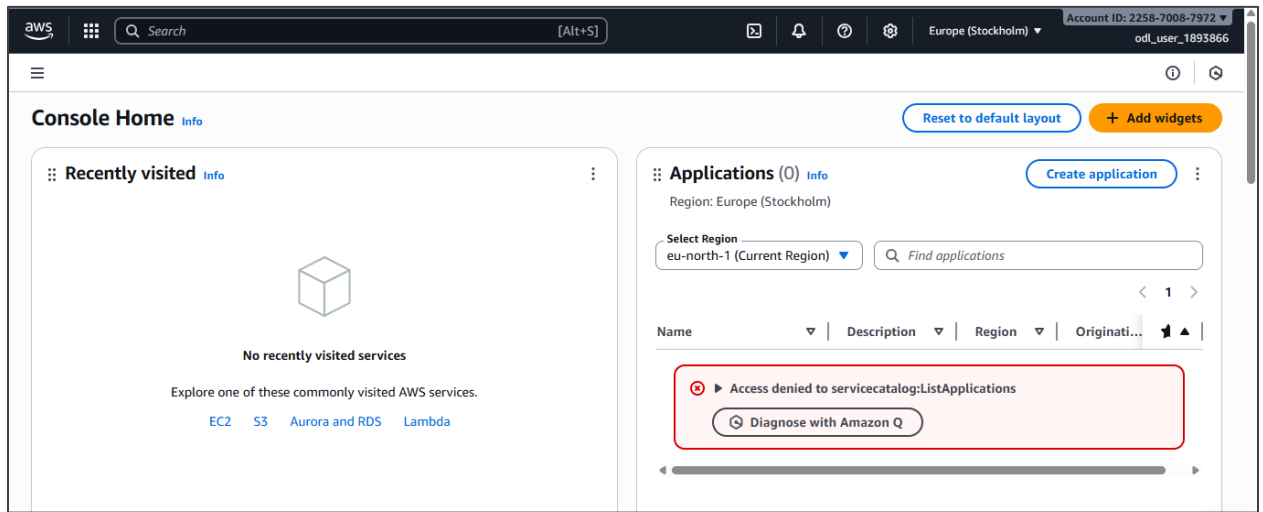
Sign in

Amazon Lightsail

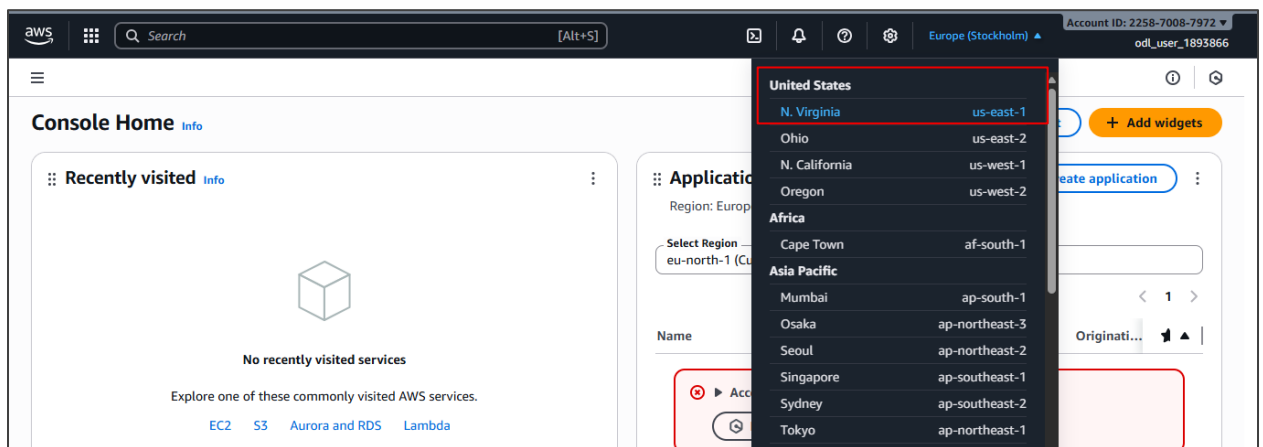
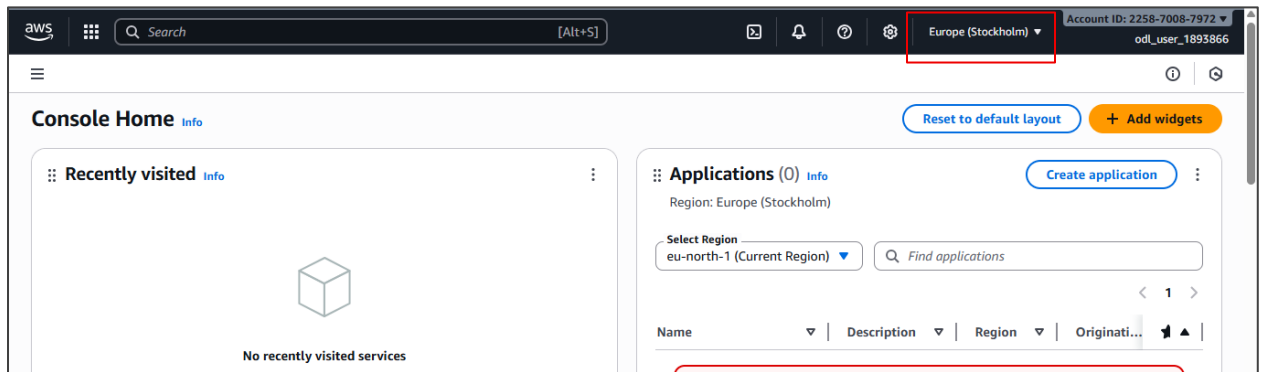
Lightsail is the easiest way to get started on AWS

Learn more »

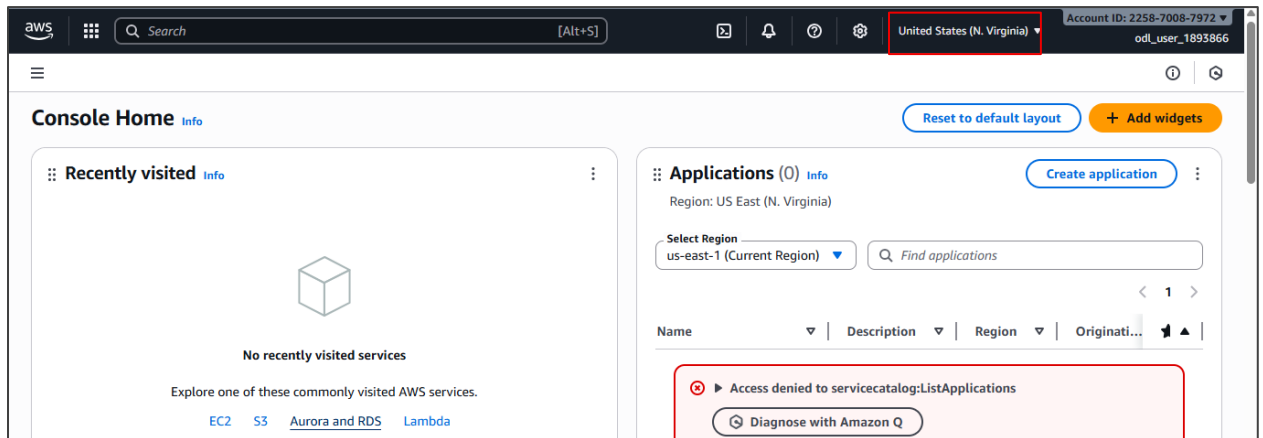
The following page appears:



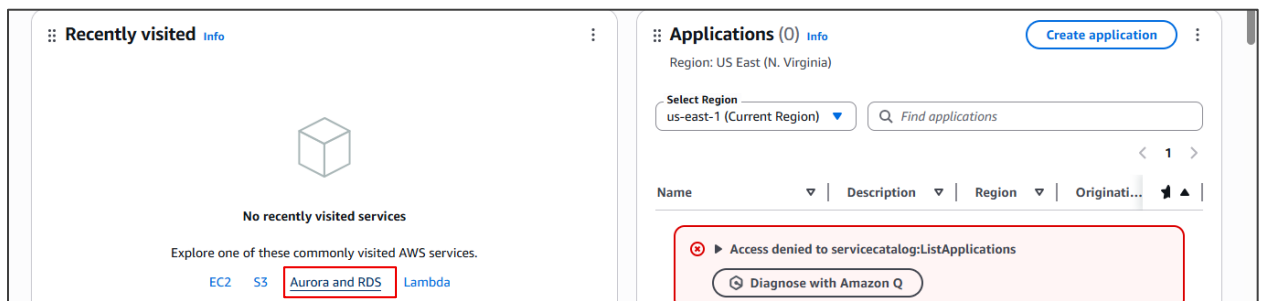
1.6 Switch the AWS region from **Europe (Stockholm)** to **United States (N. Virginia us-east-1)** using the region dropdown in the console



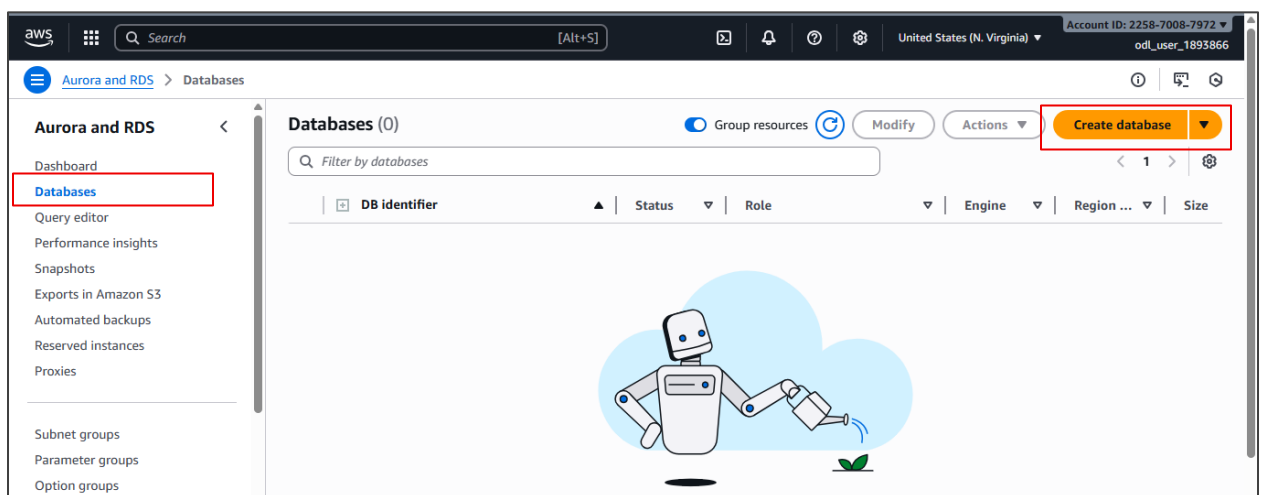
The following page appears:



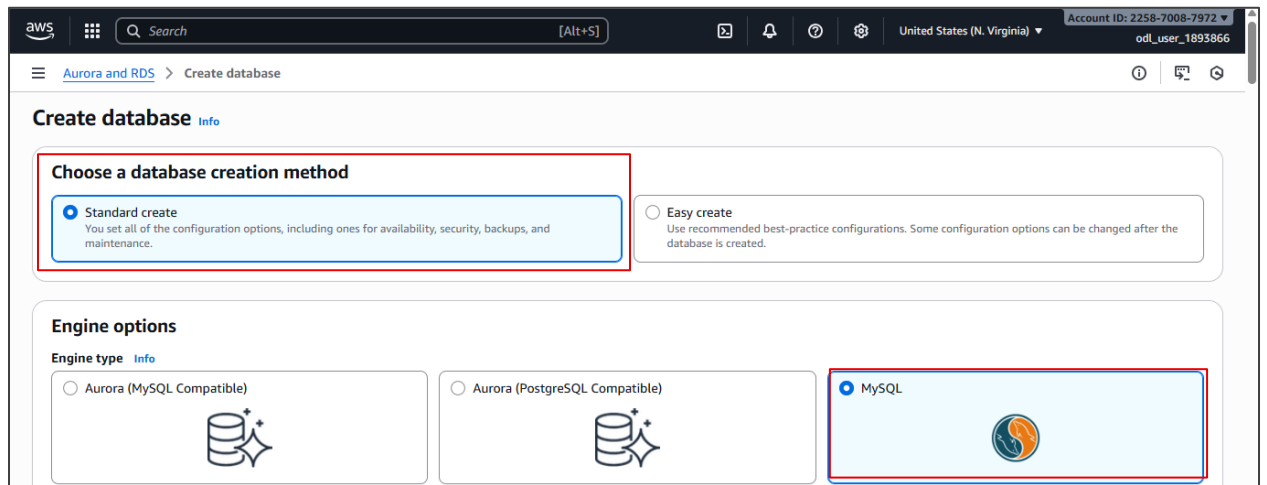
1.7 Click Aurora and RDS



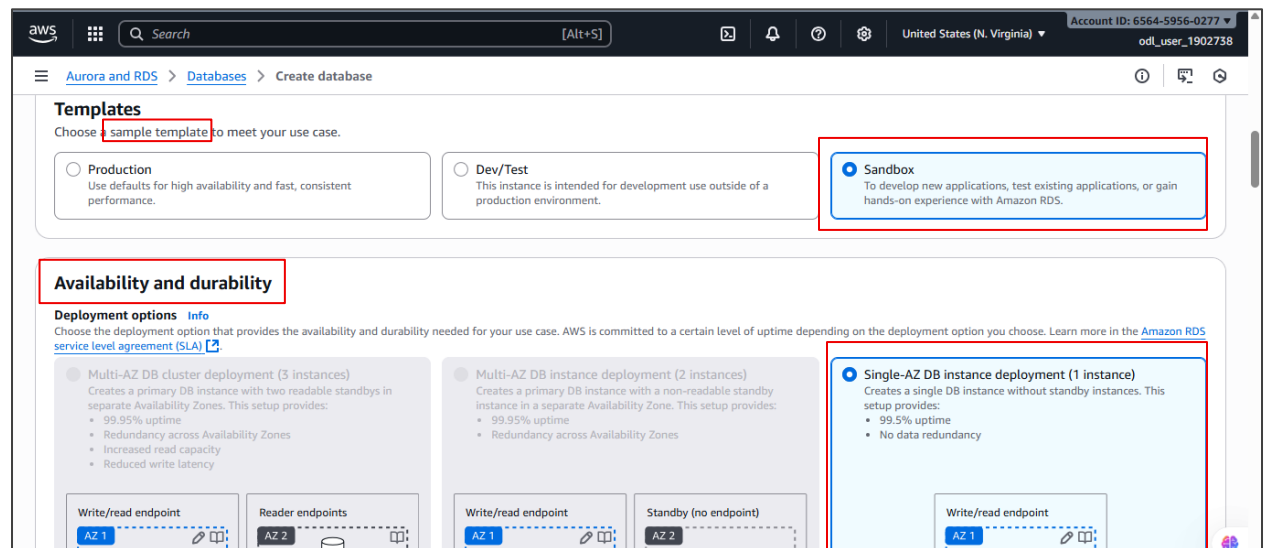
1.8 Click Databases and then Create database



1.9 Select **Standard create** as the database creation method, and choose **MySQL** as the engine type



1.10 Select **Sandbox** as the **sample template** and choose **Single-AZ DB instance deployment (1 instance)** under **Availability and durability**



- 1.11 Enter **sourceDB** as the DB instance identifier, **admin** as the Master username, and select Self managed for Credentials management

Settings

DB instance identifier [Info](#)
Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.
sourceDB
The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)
Type a login ID for the master user of your DB instance.
admin
1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - most secure
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed
Create your own password or have RDS create a password that you manage.

- 1.14 Enter a strong Master password, confirm it, and ensure the password strength indicator shows **Very strong**

Master password [Info](#)

Password strength **Very strong**
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password [Info](#)

- 1.15 Select **Burstable classes** and choose **db.t3.micro** as the DB instance class

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class [Info](#)

▼ Hide filters

☒ Show instance classes that support Amazon RDS Optimized Writes [Info](#)
Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

db.t3.micro
2 vCPUs 1 GiB RAM Network: Up to 2,085 Mbps

1.16 Select **General Purpose SSD (gp3)** as the Storage type and allocate **20 GB** as the **Allocated storage**

Storage

Storage type Info
Provisioned IOPS SSD (io2) storage volumes are now available.
General Purpose SSD (gp3)
Performance scales independently from storage

Allocated storage Info
20 GIB
Minimum: 20 GIB, Maximum: 6,144 GIB

1.17 Enable **Public access**, enable **Create new**, and name it as **sg_database**

Public access Info
☒ Yes
RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.
☐ No
RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) Info
Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.
☐ Choose existing
Choose existing VPC security groups
☒ Create new
Create new VPC security group

New VPC security group name
sg_database

RDS Proxy
RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.
☐ Create an RDS Proxy Info
RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#).

1.18 Review the configuration settings, and click **Create database** to launch the RDS instance

Log exports
Select the log types to publish to Amazon CloudWatch Logs.
☐ Audit log
☐ Error log
☐ General log
☐ iam-db-auth-error log
☐ Slow query log

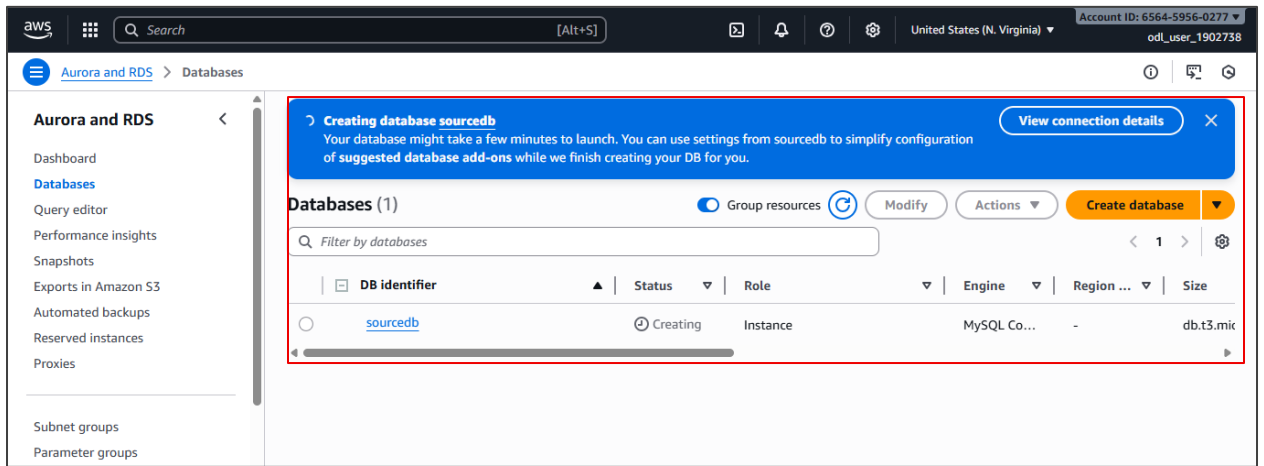
IAM role
The following service-linked role is used for publishing logs to CloudWatch Logs.
RDS service-linked role

Additional configuration
Database options, encryption turned on, backup turned on, backtracking turned off, maintenance, CloudWatch Logs, delete protection turned off.

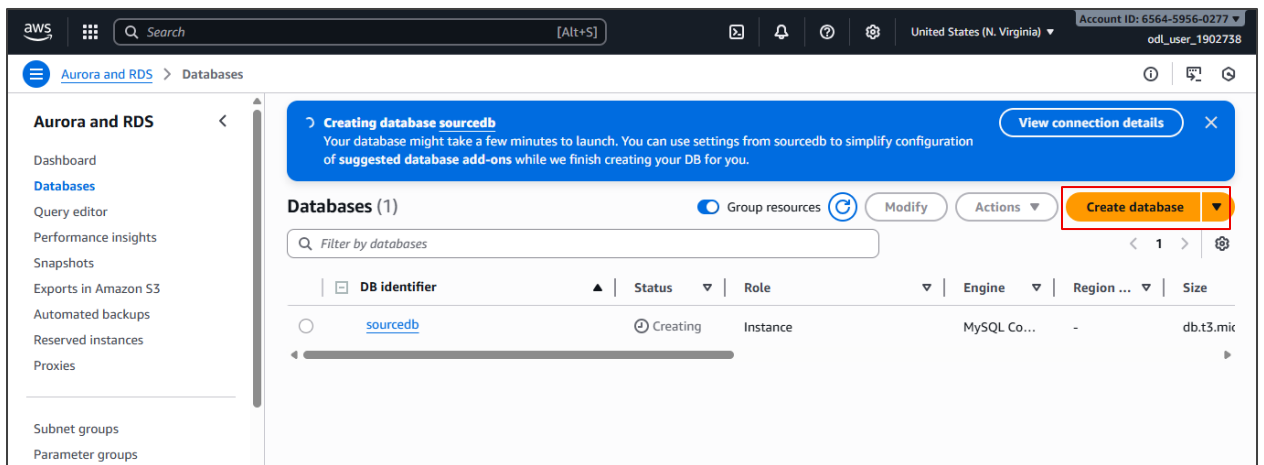
ⓘ You are responsible for ensuring that you have all of the necessary rights for any third-party products or services that you use with AWS services.

[Cancel](#) **Create database**

The following page appears:



1.19 Click **Create database**



- 1.20 Choose **Standard create** as the database creation method and **select Aurora (MySQL Compatible)** as the engine type

The screenshot shows the AWS Management Console interface for creating a database. The breadcrumb navigation at the top indicates the path: Aurora and RDS > Databases > Create database. The page title is 'Create database'. There are two main sections: 'Choose a database creation method' and 'Engine options'. In the first section, 'Standard create' is selected with a radio button, and its description is 'You set all of the configuration options, including ones for availability, security, backups, and maintenance.' The 'Easy create' option is unselected, with the description 'Use recommended best-practice configurations. Some configuration options can be changed after the database is created.' In the 'Engine options' section, 'Aurora (MySQL Compatible)' is selected with a radio button and an icon of a database cylinder with a star. The other two options, 'Aurora (PostgreSQL Compatible)' and 'MySQL', are unselected.

- 1.21 Select the sample template as **Dev/Test**, and enter **targetDB** as the DB cluster identifier

The screenshot shows the 'Templates' and 'Settings' sections of the 'Create database' page. In the 'Templates' section, 'Dev/Test' is selected with a radio button, and its description is 'This instance is intended for development use outside of a production environment.' The 'Production' template is unselected. In the 'Settings' section, the 'DB cluster identifier' field is highlighted with a red box. The text 'targetDB' is entered in this field. Below the field, there is a note: 'The DB cluster identifier is case-insensitive, but is stored as all lowercase (as in "mydbcluster"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.'

1.22 Select **Self managed** for the Credentials management, enter a strong **Master password**, confirm it, and ensure the password strength indicator shows **Very strong**

Credentials management
You can use AWS Secrets Manager or manage your master user credentials.

☐ Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ **Self managed**
Create your own password or have RDS create a password that you manage.

☐ Auto generate password
Amazon RDS can generate a password for you, or you can specify your own password.

Master password | Info

Password strength **Very strong**
Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' * @

Confirm master password | Info

1.23 Select **Aurora Standard** for Configuration options

Cluster storage configuration | Info
Choose the storage configuration for the Aurora DB cluster that best fits your application's price predictability and price performance needs.

Configuration options

☐ Aurora I/O-Optimized

- Predictable pricing for all applications. Improved price performance for I/O-intensive applications (I/O costs >25% of total database costs).
- No additional charges for read/write I/O operations. DB instance and storage prices include I/O usage.

☒ **Aurora Standard**

- Cost-effective pricing for many applications with moderate I/O usage (I/O costs <25% of total database costs).
- Pay-per-request I/O charges apply. DB instance and storage prices don't include I/O usage.

1.24 Select **Burstable classes**, and choose **db.t3.micro** as DB instance class

Instance configuration
The DB instance configuration options below are limited to those supported by the engine that you selected above.

DB instance class | Info

▼ Hide filters

☐ Include previous generation classes

☐ Serverless v2

☐ Memory optimized classes (includes r classes)

☒ **Burstable classes (includes t classes)**

db.t3.medium
2 vCPUs 4 GiB RAM Network: Up to 2,085 Mbps

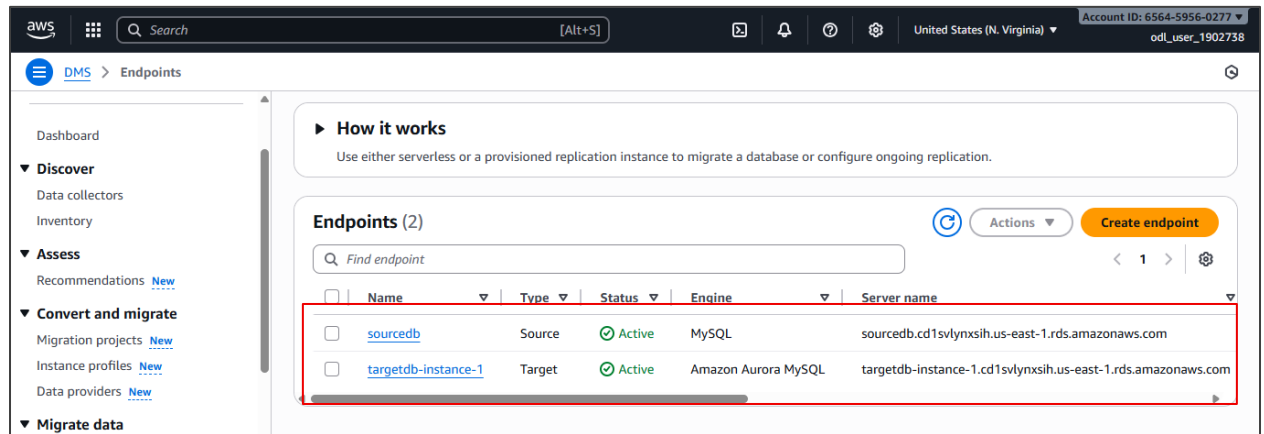
1.25 Enable **Public access**, **Choose existing**, and select **sg_database** from the **Existing VPC security groups**

The screenshot shows the 'Create database' page in the AWS Management Console. The 'Public access' section has 'Yes' selected. The 'VPC security group (firewall)' section has 'Choose existing' selected, and 'sg_database' is chosen from the 'Existing VPC security groups' dropdown. The 'Availability Zone' is set to 'No preference'.

1.26 Scroll down and click **Create database**

The screenshot shows the 'Create database' page in the AWS Management Console, scrolled down to the bottom. The 'Additional configuration' section is visible, and the 'Create database' button is highlighted with a red box.

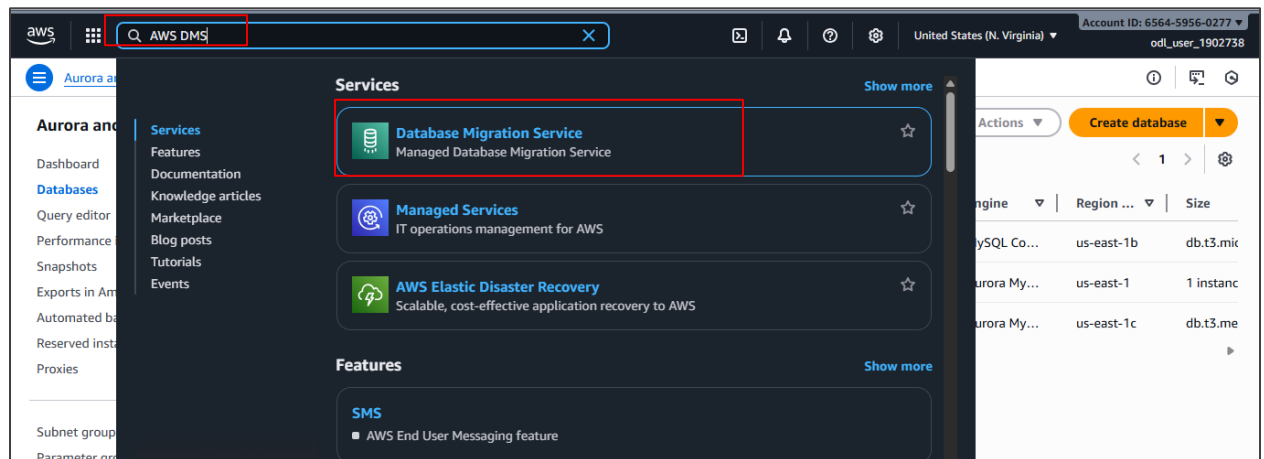
The following page appears:



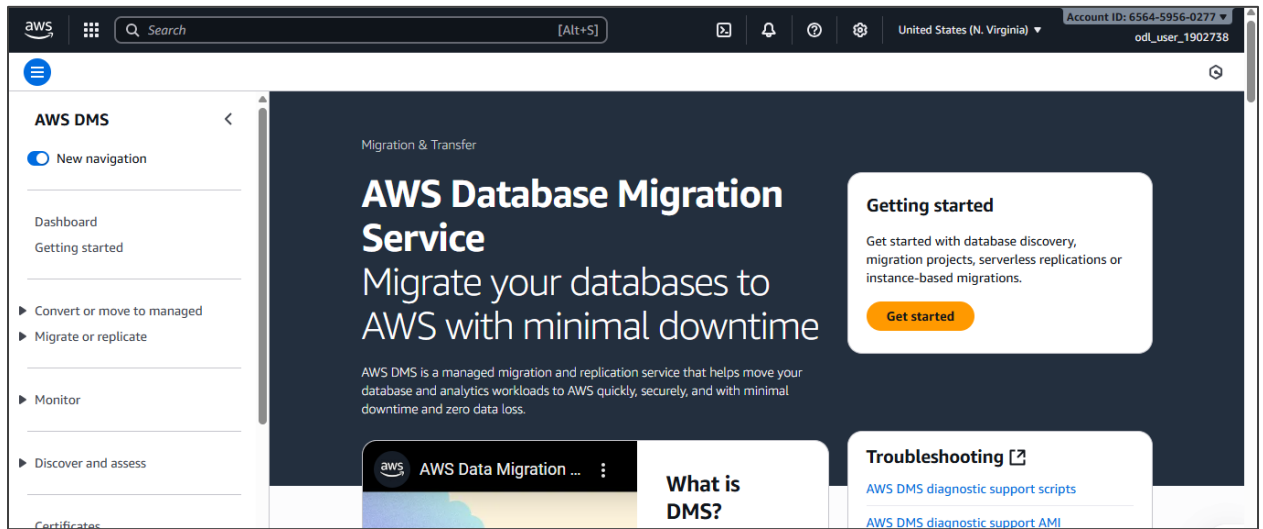
Note: Record the server names of both the source and target; we will use them in the upcoming steps.

Step 2: Create a replication instance in AWS DMS

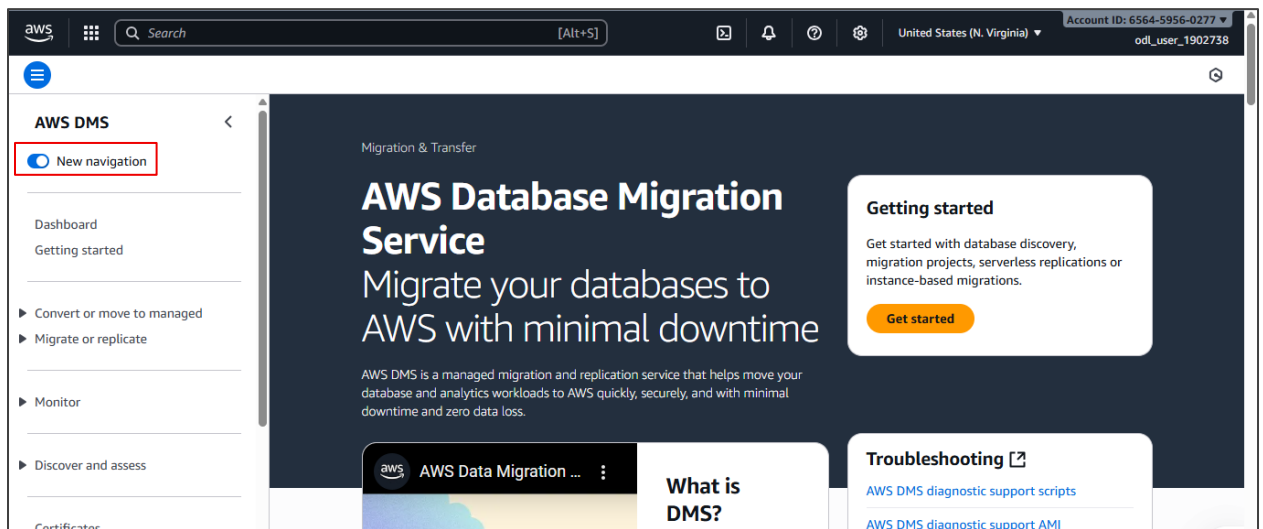
2.1 Search for **AWS DMS** in the AWS console search bar and select **Database Migration Service** from the services list



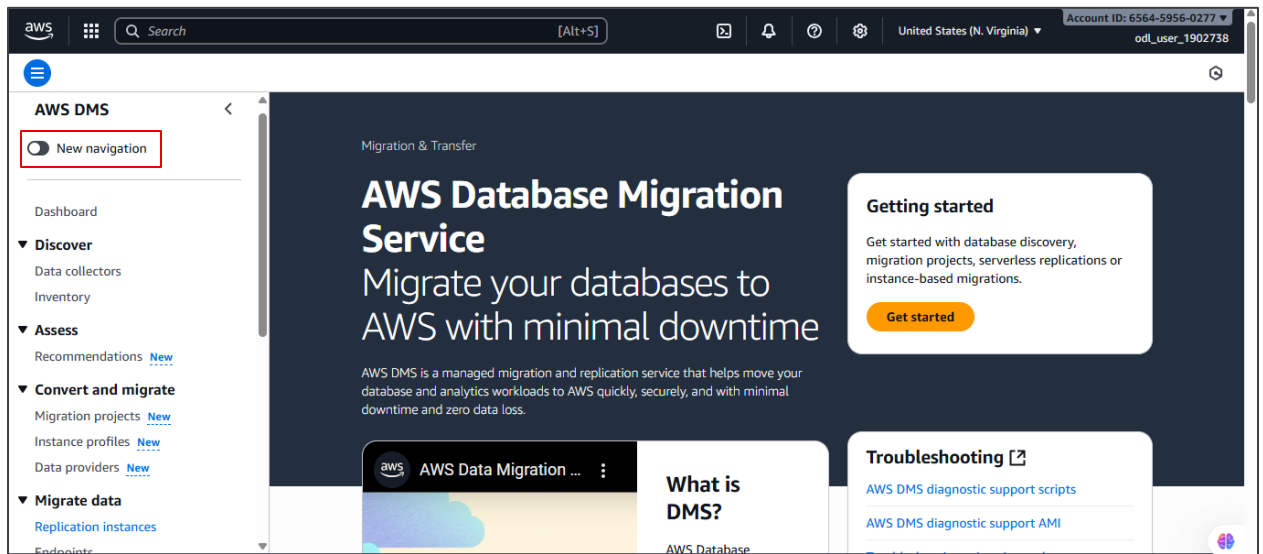
The following page opens:



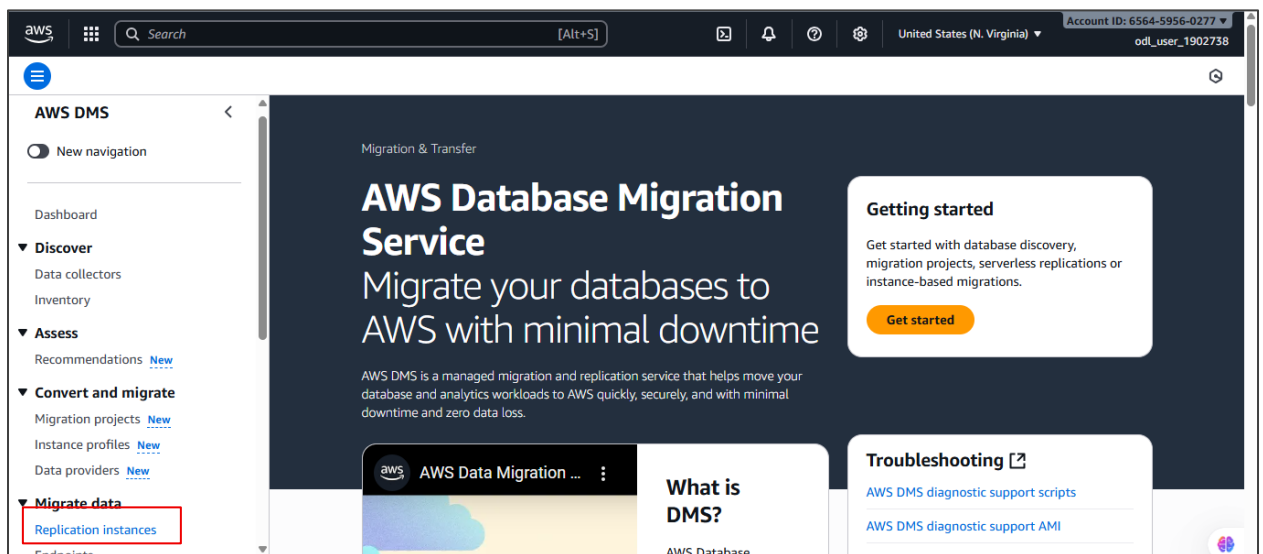
2.2 Turn off the New navigation button



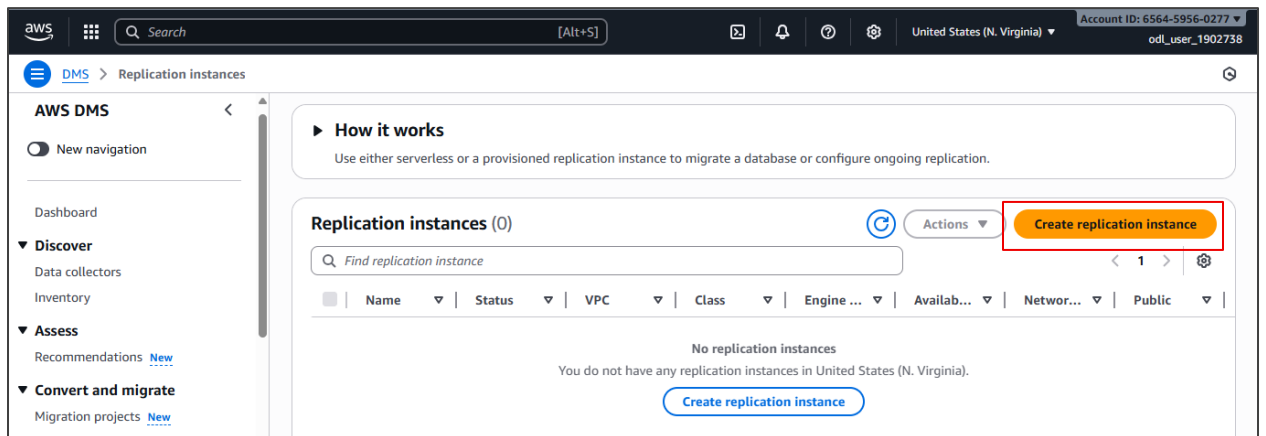
The following page appears:



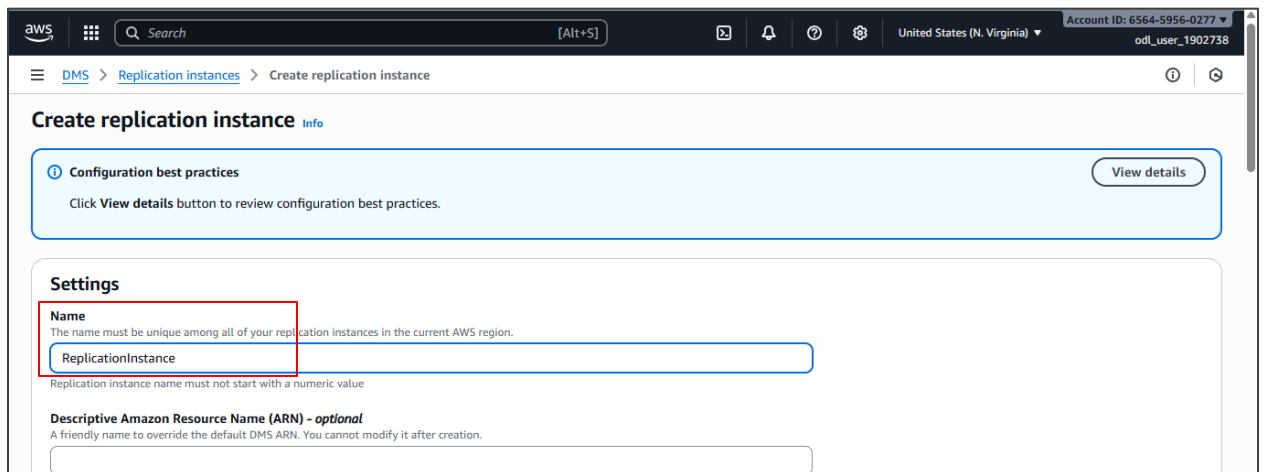
2.3 Click Replication instances



2.4 Click **Create replication instance**



2.5 Enter **ReplicationInstance** in the Name field



2.6 Select **Dev or test workload (Single-AZ)** for the **High Availability** option and choose **IPv4** for **Network type**

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Replication instances > Create replication instance

High Availability Info
The Multi-AZ option deploys a primary replication instance in one Availability Zone (AZ) and a standby in another AZ. The Single-AZ option deploys a single replication instance in one AZ. Billing is based on DMS pricing.

Dev or test workload (Single-AZ) ▼

Storage Info
Allocated storage (GiB)
Choose the amount of storage space you want for your replication instance. AWS DMS uses this storage for log files and cached transactions while replication tasks are in progress.

50

Connectivity and security Info
Network type - new Info
To use dual-stack mode, make sure that you associate an IPv6 CIDR block with a subnet in the VPC you specify.

☒ **IPv4**
Replication instance with an IPv4 network type that supports IPv4 addressing.

☐ **Dual-stack mode**
Replication instance with a dual network type that supports both IPv4 and IPv6 addressing.

2.7 Scroll down and click **Create replication instance**

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Replication instances > Create replication instance

► **Kerberos authentication - optional**
Choose Kerberos authentication to provide secure access to your database endpoints. You can use this authentication mechanism for DMS version 3.5.3 and higher and supported database engines. [Learn more](#)

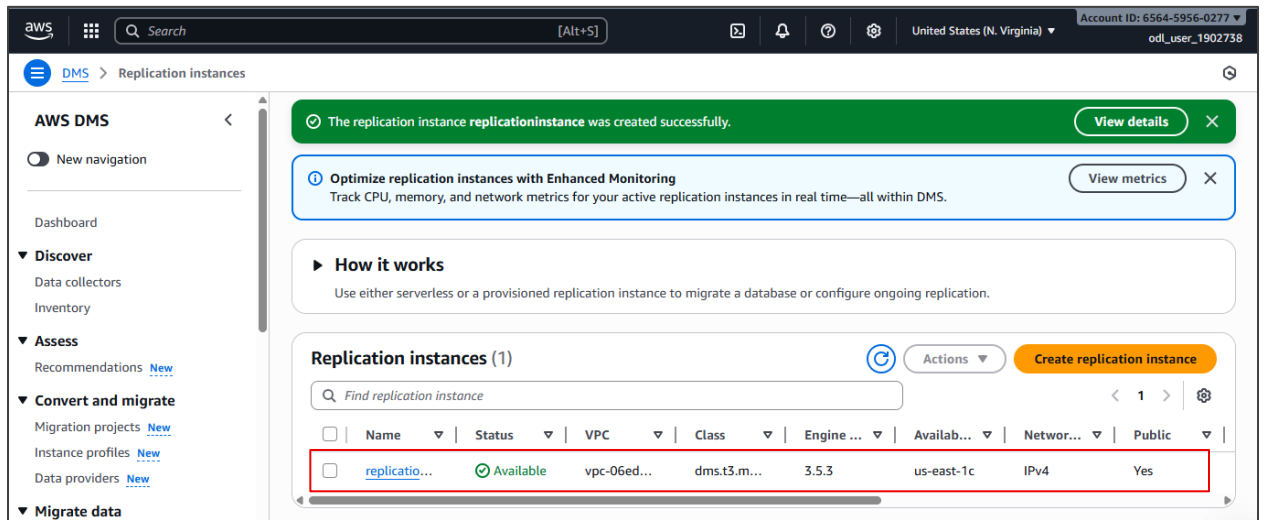
► **Tags - optional**
Add tags to organize your DMS resources. You can use tags to manage your IAM roles and policies, and track your DMS costs.

ⓘ The required IAM roles **dms-vpc-role**, **dms-cloudwatch-logs-role** do not exist. The **dms-vpc-role** gives DMS the permission to use AWS resources to run replication instance within specified VPC. The **dms-cloudwatch-logs-role** gives DMS the permission to publish logs to Amazon CloudWatch. When you choose **Create replication instance**, DMS automatically creates these roles with the required permissions for you.

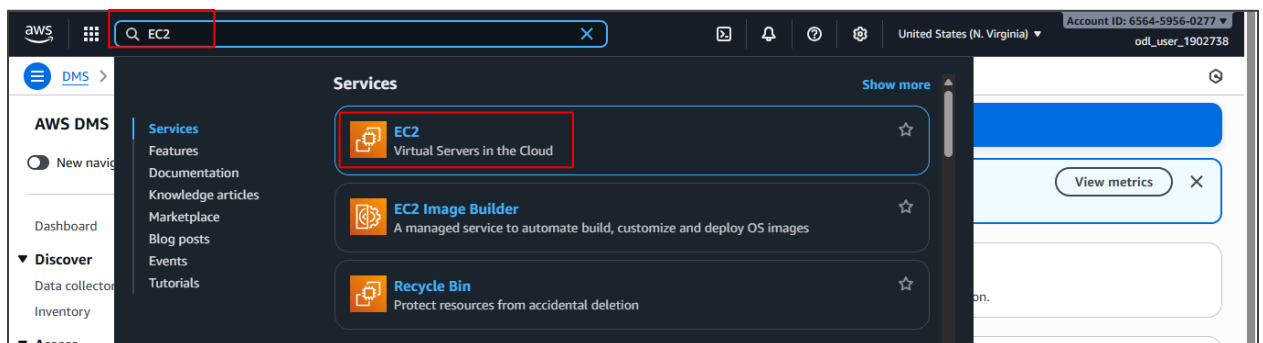
- Check [permissions](#) for dms-vpc-role
- Check [permissions](#) for dms-cloudwatch-logs-role

Cancel **Create replication instance**

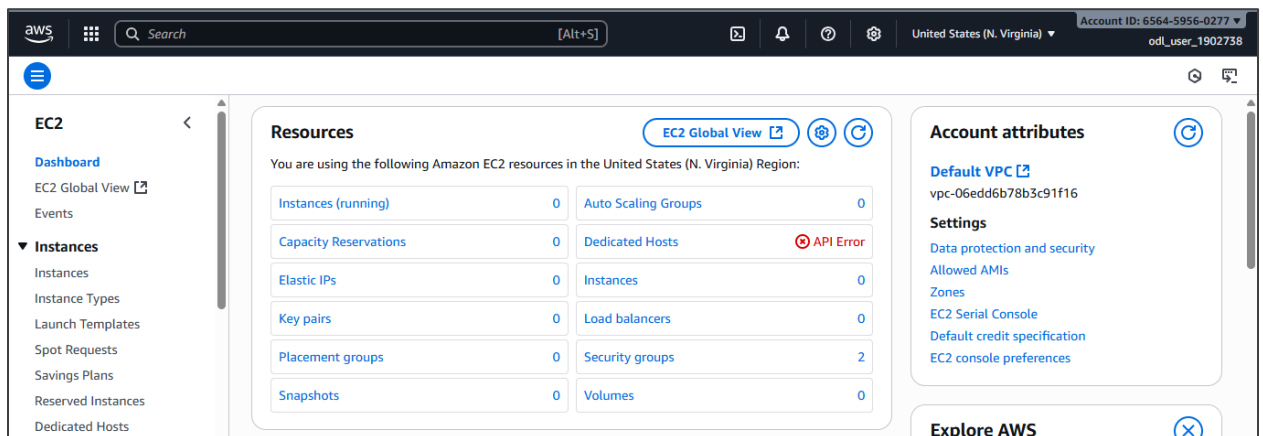
The following page appears:



2.8 Search for **EC2** in the AWS console search bar and click **EC2 Virtual Servers in the Cloud** from the available options



The following page appears:



2.9 Click Security groups

The screenshot shows the AWS Management Console interface. On the left, the 'EC2' menu is expanded, showing 'Dashboard', 'EC2 Global View', 'Events', and 'Instances'. The 'Resources' section in the center displays a grid of EC2 resources. The 'Security groups' link is highlighted with a red box. The 'Account attributes' section on the right shows the 'Default VPC' and 'Settings'.

Resource	Count
Instances (running)	0
Capacity Reservations	0
Elastic IPs	0
Key pairs	0
Placement groups	0
Snapshots	0
Auto Scaling Groups	0
Dedicated Hosts	API Error
Instances	0
Load balancers	0
Security groups	2
Volumes	0

2.10 Click sg-01b3a53527b7c3556

The screenshot shows the 'Security Groups' page in the AWS Management Console. The 'Security Groups (2)' section is active, displaying a table of security groups. The security group 'sg-01b3a53527b7c3556' is highlighted with a red box. The table includes columns for Name, Security group ID, Security group name, and VPC ID.

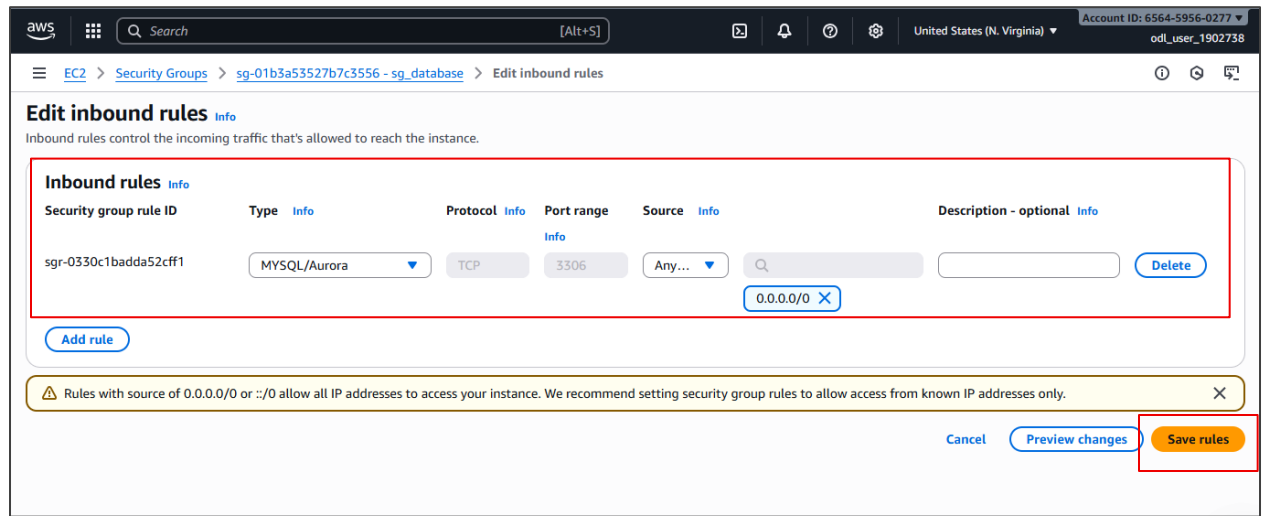
Name	Security group ID	Security group name	VPC ID
-	sg-0ced2c7e71c4c0359	default	vpc-06edd6b78b3c91f16
-	sg-01b3a53527b7c3556	sg_database	vpc-06edd6b78b3c91f16

2.11 Click Edit inbound rules

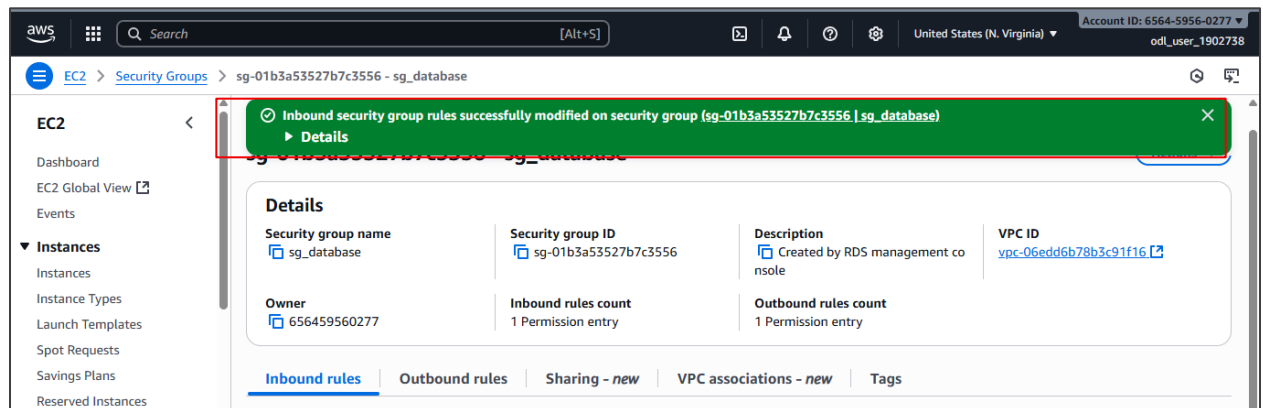
The screenshot shows the 'sg-01b3a53527b7c3556 - sg_database' page in the AWS Management Console. The 'Inbound rules' tab is selected, showing a table of inbound rules. The 'Edit inbound rules' button is highlighted with a red box. The table includes columns for Name, Security group rule ID, IP version, Type, Protocol, and Port.

Name	Security group rule ID	IP version	Type	Protocol	Port
-	sg-0330c1badda52cfe1	IPv4	MySQL/Aurora	TCP	3306

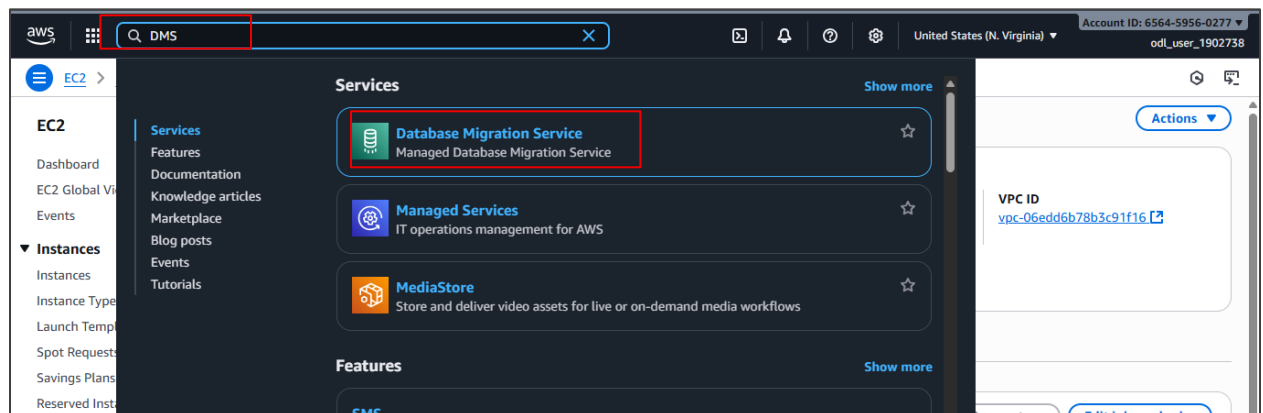
2.12 Select **MySQL/Aurora** as the Type, confirm that **TCP/3306** is auto-filled, choose **Anywhere IPv4** as the Source, and click **Save rules**



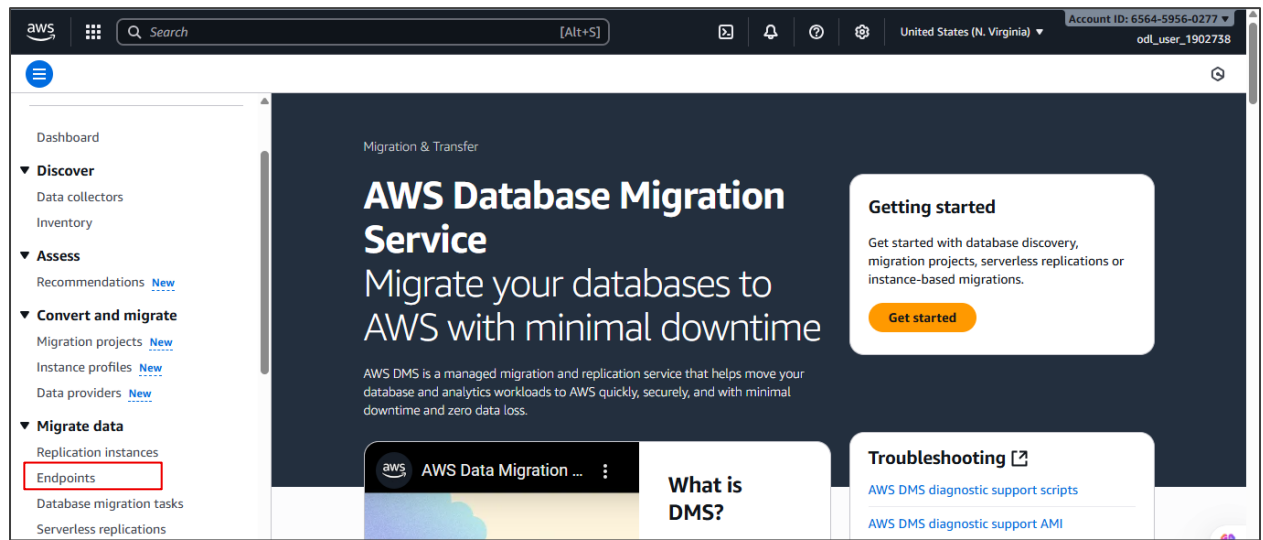
The following page appears:



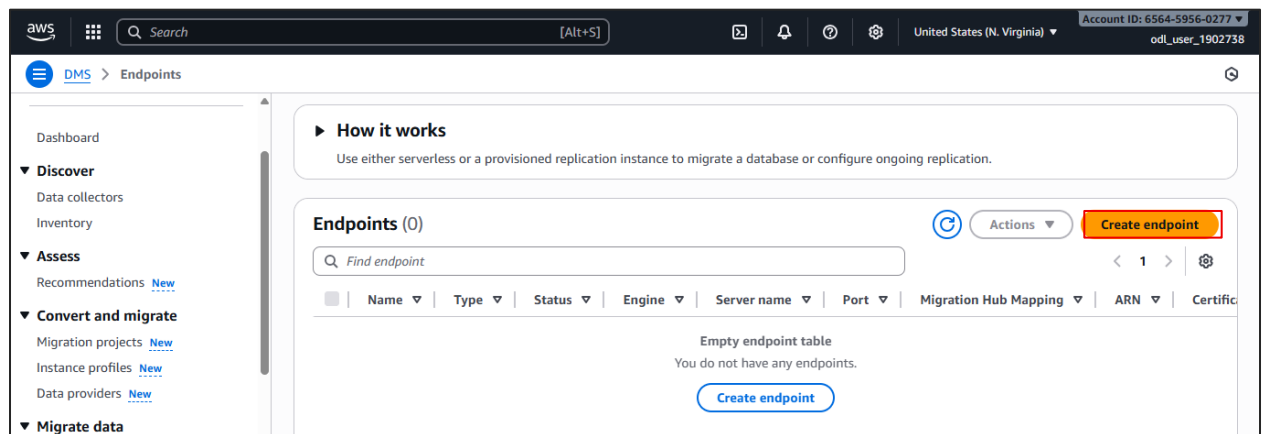
2.13 Search for **DMS** in the search bar and select **Database Migration Service**



2.14 Click Endpoints



2.15 Click Create endpoint



2.16 Choose **Source endpoint**, check **Select RDS DB instance**, choose your RDS instance (**sourcedb**), and set the Endpoint identifier as **sourcedb**

The screenshot shows the AWS DMS 'Create endpoint' page. The 'Endpoint type' section has two options: 'Source endpoint' (selected) and 'Target endpoint'. The 'Source endpoint' description states: 'A source endpoint allows AWS DMS to read data from a database (on-premises or in the cloud), or from other data source such as Amazon S3.' The 'Target endpoint' description states: 'A target endpoint allows AWS DMS to write data to a database, or to other data stores such as Amazon DynamoDB or Kinesis.' Below this, the 'Select RDS DB instance' checkbox is checked. The text below it says: 'Choose this option if the endpoint is an Amazon RDS DB instance. It provides a list of available RDS Instances from the current region.' The 'RDS Instance' dropdown menu is open, showing 'sourcedb' as the selected instance. The 'Endpoint configuration' section has the 'Endpoint identifier' field set to 'sourcedb'.

2.17 Select **MySQL** for the **Source engine** and enable **Provide access information manually**

The screenshot shows the AWS DMS 'Create endpoint' page. The 'Source engine' dropdown menu is set to 'MySQL'. The 'Access to endpoint database' section has three options: 'AWS Secrets Manager', 'Provide access information manually' (selected), and 'IAM authentication - new'. The 'Provide access information manually' option has a red box around it. The 'Server name' field is set to 'sourcedb.cd1svlynxsih.us-east-1.rds.amazonaws.com' and the 'Port' field is set to '3306'.

2.18 Enter the **User name** as **admin** and give a strong **Password**

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Endpoints > Create endpoint

3306

User name | Info Password | Info

admin

Secure Socket Layer (SSL) mode | Info

The type of Secure Socket Layer enforcement

none

2.19 Scroll down and click **Test endpoint connection**

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Endpoints > Create endpoint

Secure Socket Layer (SSL) mode | Info

The type of Secure Socket Layer enforcement

none

Endpoint settings

KMS key

Tags

Add tags to your DMS resources to organize and track your DMS costs.

Test endpoint connection - optional

Choose the replication instance to test the network and database connectivity for migration.

Cancel Create endpoint

2.20 Click **Run test**

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Endpoints > Create endpoint

Test endpoint connection - optional

Choose the replication instance to test the network and database connectivity for migration.

Important: Connection testing for AWS DMS Serverless endpoints

When using endpoints with DMS Serverless replications, you cannot perform connection testing through this interface. DMS Serverless automatically validates your endpoint connection as part of the startup sequence when you start replication. [Learn more](#)

Replication instance

A replication instance performs the database migration.

replicationinstance

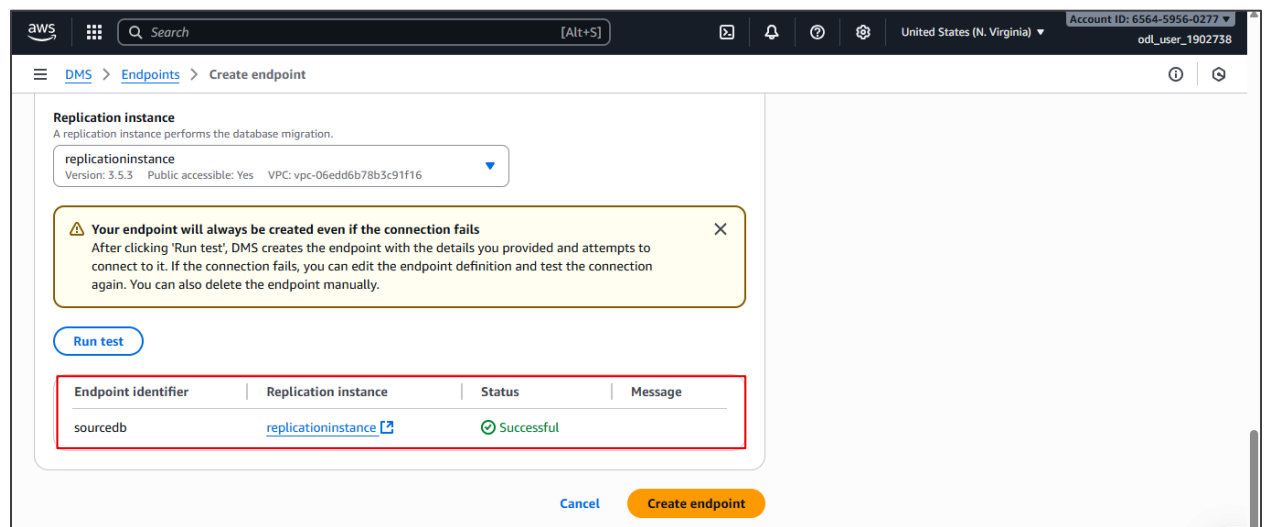
Version: 3.5.3 Public accessible: Yes VPC: vpc-06edd6b78b3c91f16

Your endpoint will always be created even if the connection fails

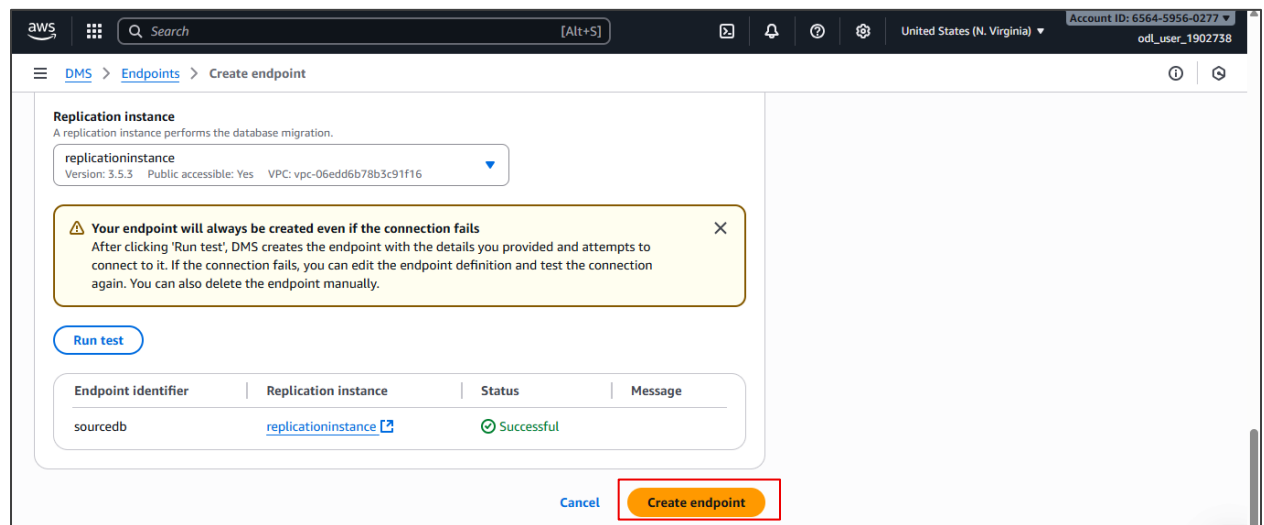
After clicking 'Run test', DMS creates the endpoint with the details you provided and attempts to connect to it. If the connection fails, you can edit the endpoint definition and test the connection again. You can also delete the endpoint manually.

Run test

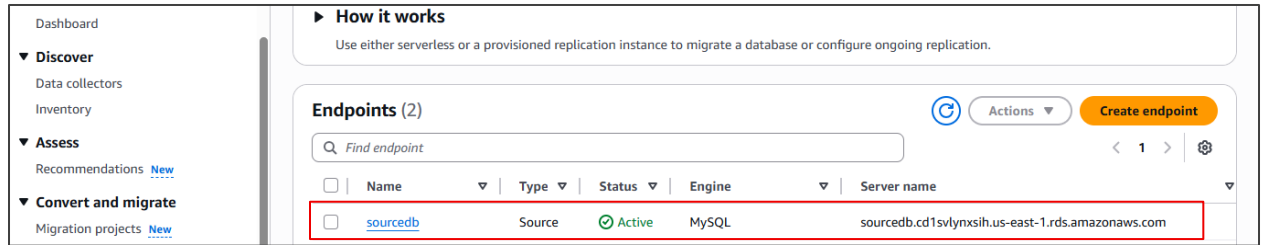
The following page appears:



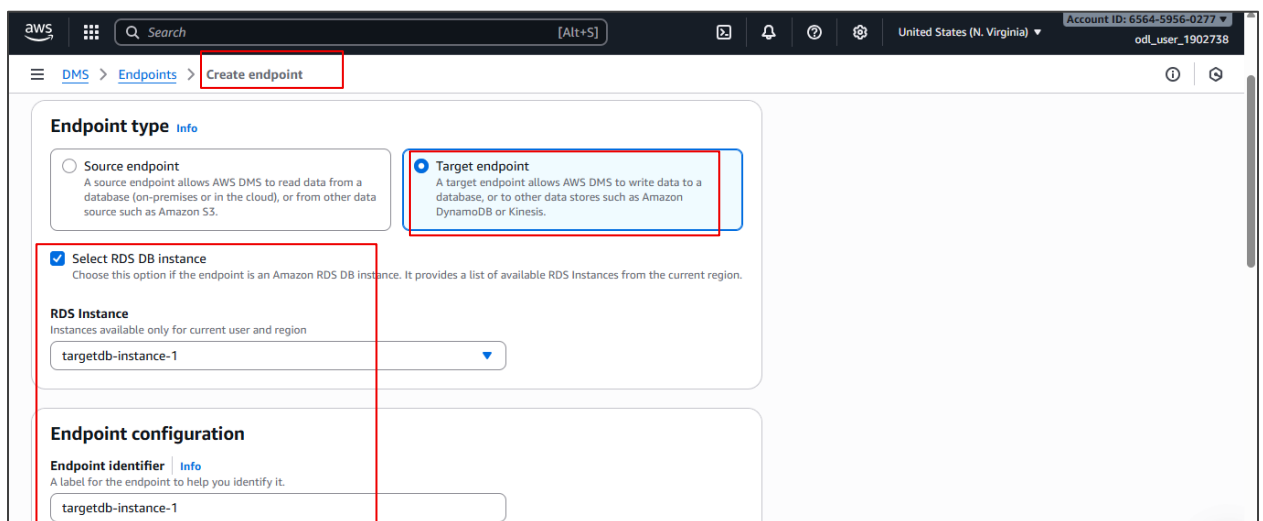
2.21 Click **Create endpoint**



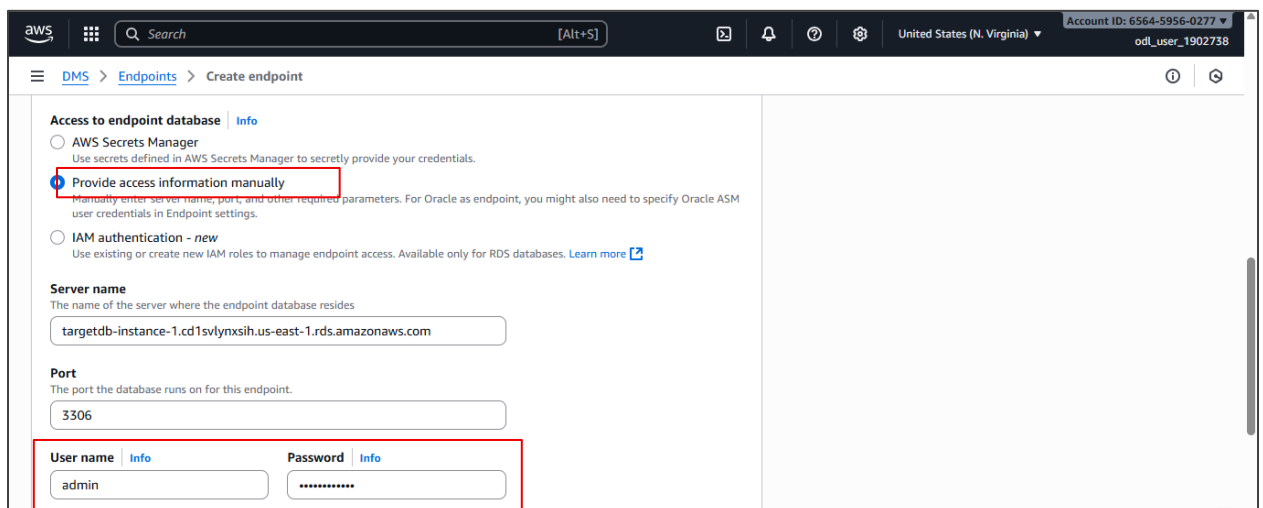
The following page appears:



- 2.22 Navigate to **Create endpoint**, choose **Target endpoint**, check **Select RDS DB instance**, choose your RDS instance (**targetdb-instance-1**), and set the Endpoint identifier as **targetdb-instance-1**



- 2.23 Enable the **Provide access information manually** option, enter the **User name** as **admin**, and enter the password used in step 2.18



Note: Use the same password throughout the demo

2.24 Scroll down and click **Test endpoint connection**

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Endpoints > Create endpoint

Secure Socket Layer (SSL) mode [Info](#)
The type of Secure Socket Layer enforcement

none

▶ Endpoint settings

▶ KMS key

▶ Tags
Add tags to your DMS resources to organize and track your DMS costs.

▶ **Test endpoint connection - optional**
Choose the replication instance to test the network and database connectivity for migration.

Cancel Create endpoint

2.25 Click **Run test**

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Endpoints > Create endpoint

Replication instance
A replication instance performs the database migration.

replicationinstance
Version: 3.5.3 Public accessible: Yes VPC: vpc-06edd6b78b3c91f16

⚠ **Your endpoint will always be created even if the connection fails**
After clicking 'Run test', DMS creates the endpoint with the details you provided and attempts to connect to it. If the connection fails, you can edit the endpoint definition and test the connection again. You can also delete the endpoint manually.

Run test

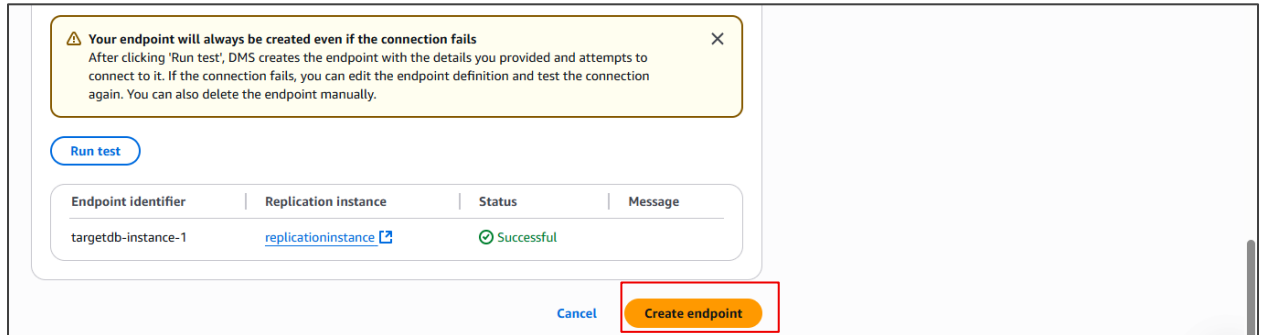
The following page appears:

Run test

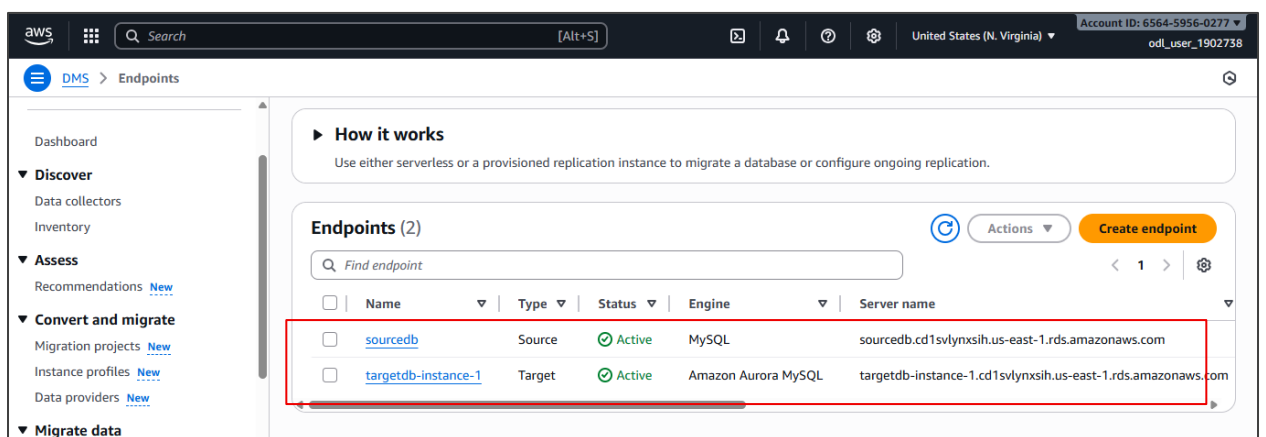
Endpoint identifier	Replication instance	Status	Message
targetdb-instance-1	replicationinstance	Successful	

Cancel Create endpoint

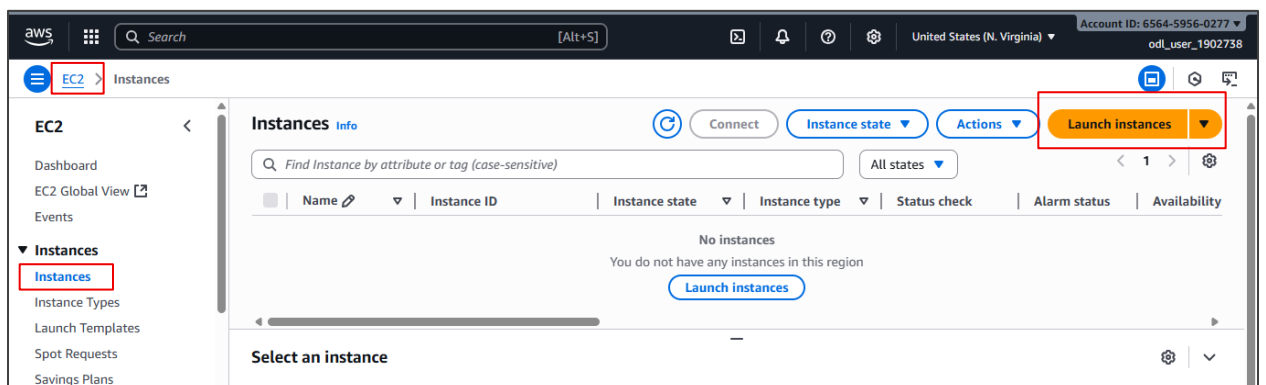
2.26 Click **Create endpoint**



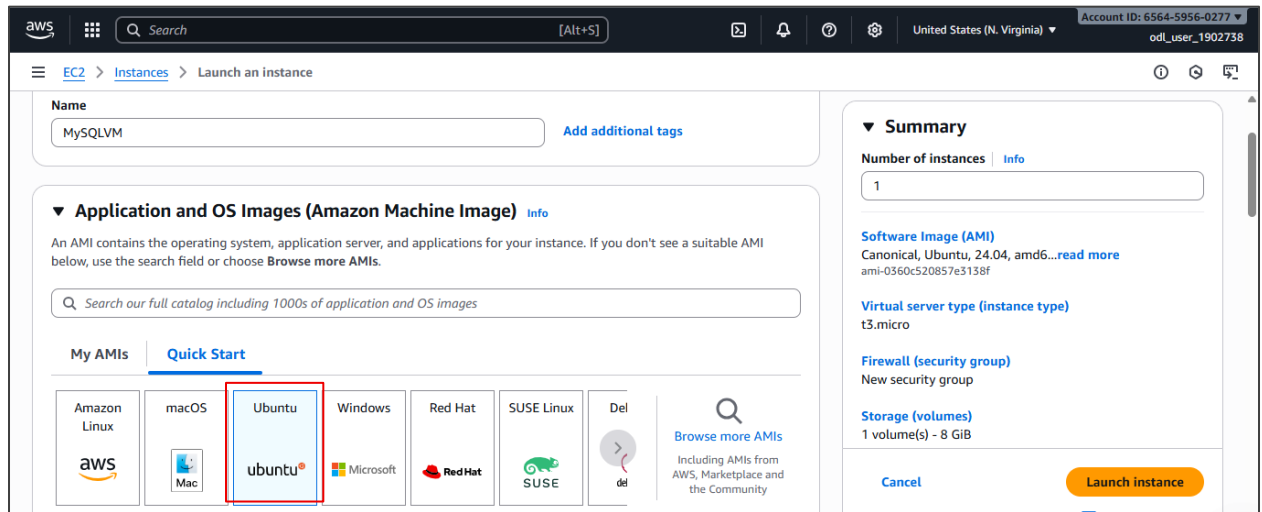
The following page appears:



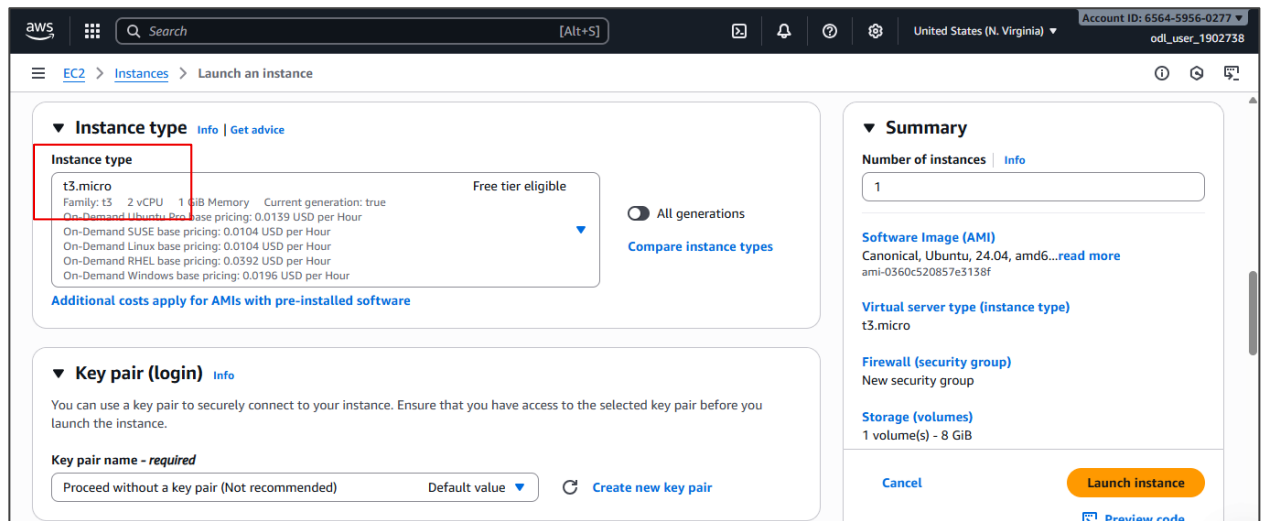
2.27 Navigate to **EC2**, click **Instances**, and then click **Launch instances**



2.28 Select Ubuntu



2.29 Select t3.micro as Instance type



2.30 Scroll down and click Launch instance

▼ Configure storage Info

Advanced

1x 8 GiB gp3

Root volume, 3000 IOPS, Not encrypted

Add new volume

The selected AMI contains instance store volumes, however the instance does not allow any instance store volumes. None of the instance store volumes from the AMI will be accessible from the instance

Click refresh to view backup information

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

0 x File systems

Edit

Number of instances 1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-0560c520857e3138f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel

Launch instance

The following page appears:

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 6564-5956-0277

odl_user_1902738

EC2

Instances

Launch an instance

Success

Successfully initiated launch of instance (i-01dc40c01baccdb44)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

Connect an RDS database

Create EBS snapshot policy

Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

2.31 Click i-01dc40c01baccdb44

aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 6564-5956-0277

odl_user_1902738

EC2

Instances

Launch an instance

Success

Successfully initiated launch of instance (i-01dc40c01baccdb44)

Launch log

Next Steps

What would you like to do next with this instance, for example "create alarm" or "create backup"

Create billing usage alerts

To manage costs and avoid surprise bills, set up email notifications for billing usage thresholds.

Create billing alerts

Connect to your instance

Once your instance is running, log into it from your local computer.

Connect to instance

Connect an RDS database

Configure the connection between an EC2 instance and a database to allow traffic flow between them.

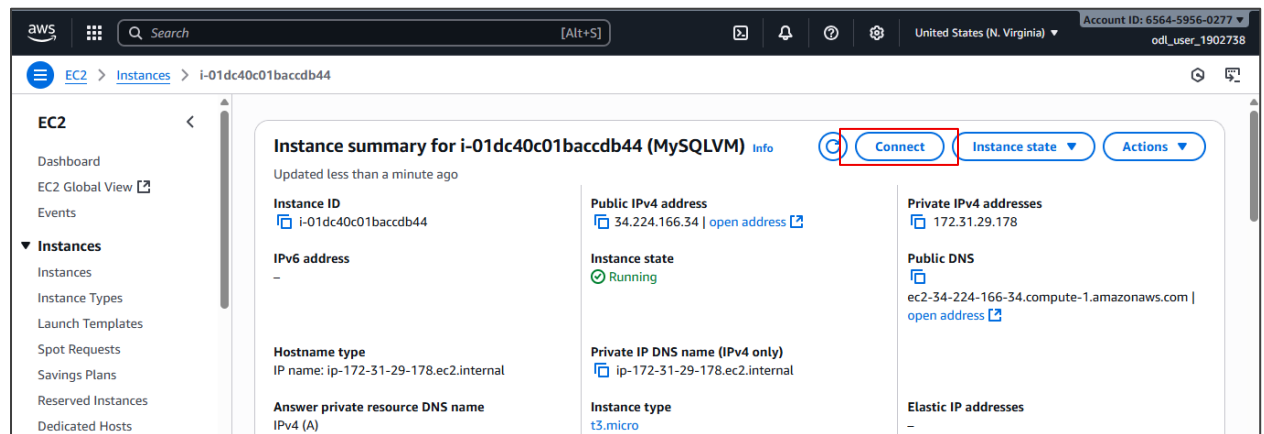
Connect an RDS database

Create EBS snapshot policy

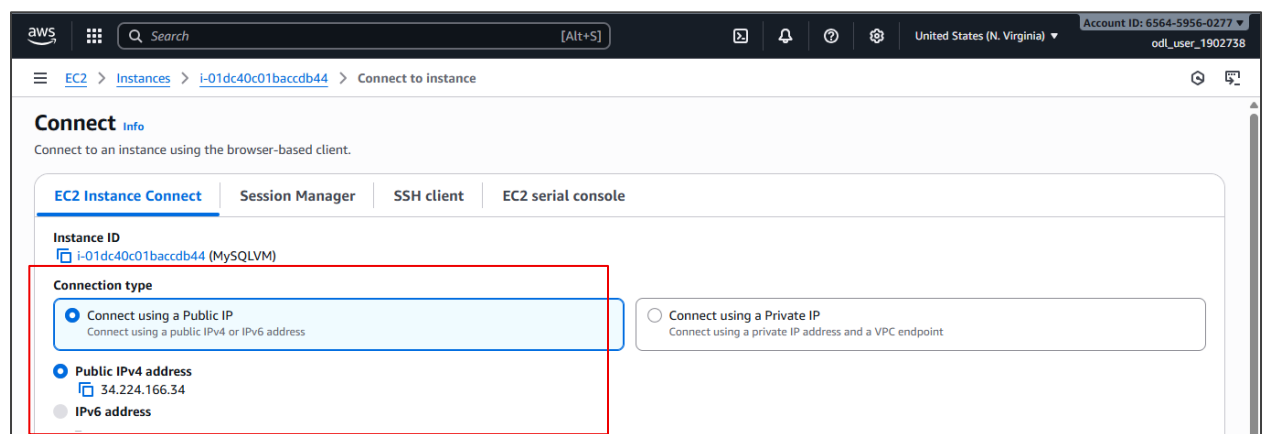
Create a policy that automates the creation, retention, and deletion of EBS snapshots

Create EBS snapshot policy

2.32 Click **Connect**



2.33 Enable **Connect** using a Public IP



2.34 Click **Connect**

aws [Search] [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

EC2 > Instances > i-01dc40c01bacddb44 > Connect to instance

Instance ID
i-01dc40c01bacddb44 (MySQLVM)

Connection type

☒ Connect using a Public IP
Connect using a public IPv4 or IPv6 address

☐ Connect using a Private IP
Connect using a private IP address and a VPC endpoint

Public IPv4 address
34.224.166.34

IPv6 address
-

Username
Enter the username defined in the AMI used to launch the instance. If you didn't define a custom username, use the default username, ubuntu.
ubuntu

Note: In most cases, the default username, ubuntu, is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

Cancel **Connect**

The following page opens:

```
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-29-178:~$
```

i-01dc40c01bacddb44 (MySQLVM)
PublicIPs: 34.224.166.34 PrivateIPs: 172.31.29.178

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Step 3: Implement database migration using AWS DMS

- 3.1 Run the following command to switch to the root user:
sudo su -

```
aws [Search] [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-29-178:~$ sudo su -
root@ip-172-31-29-178:~#

i-01dc40c01baccdb44 (MySQLVM)
PublicIPs: 34.224.166.34 PrivateIPs: 172.31.29.178
```

3.2 Run the following command to update and install mysql-client:
apt update && apt install mysql-client-core-8.0

```
aws [Search] [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-29-178:~$ sudo su -
root@ip-172-31-29-178:~# apt update && apt install mysql-client-core-8.0
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease [126 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1483 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [286 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [15.3 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1486 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [300 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
```

3.3 Run the following command to connect sourcedb, and enter the admin password:
mysql -h sourcedb.cxy2somegb0m.us-east-1.rds.amazonaws.com -u admin -p

```
aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

Need to get 2740 kB of archives.
After this operation, 61.7 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 mysql-client-core-8.0 amd64 8.0.43-0ubuntu0.24.04.2 [2740 kB]
Fetched 2740 kB in 0s (42.2 MB/s)
Selecting previously unselected package mysql-client-core-8.0.
(Reading database ... 71718 files and directories currently installed.)
Preparing to unpack .../mysql-client-core-8.0_8.0.43-0ubuntu0.24.04.2_amd64.deb ...
Unpacking mysql-client-core-8.0 (8.0.43-0ubuntu0.24.04.2) ...
Setting up mysql-client-core-8.0 (8.0.43-0ubuntu0.24.04.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@ip-172-31-29-178:~# mysql -h sourcedb.cxy2somegb0m.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
```

Note: The command will change based on the server name of sourcedb obtained in step 1.26.
Format: **mysql -h <server name> -u admin -p**

3.2 Run the following SQL statements on sourcedb:

-- 1. Create a Database

```
CREATE DATABASE company_db;
```

-- 2. Use the Database

```
USE company_db;
```

-- 3. Create a Table

```
CREATE TABLE employees (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    first_name VARCHAR(50),  
    last_name VARCHAR(50),  
    department VARCHAR(50),  
    salary DECIMAL(10,2)  
);
```

-- 4. Insert Some Records

```
INSERT INTO employees (first_name, last_name, department, salary)  
VALUES  
('John', 'Doe', 'HR', 45000.00),  
('Jane', 'Smith', 'Finance', 55000.00),  
('Robert', 'Brown', 'IT', 65000.00),  
('Emily', 'Davis', 'Marketing', 50000.00),  
('Michael', 'Wilson', 'IT', 70000.00);
```

-- 5. Verify Data

SELECT * FROM employees;

```
aws [Search] [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

mysql> 1. Create a Database
-> CREATE DATABASE company_db;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '1. Create a Database
CREATE DATABASE company_db' at line 1
mysql> -- 1.Create a Database
mysql> CREATE DATABASE company_db;
Query OK, 1 row affected (0.02 sec)

mysql> -- 2.Use the database
mysql> USE company_db
Database changed
mysql> -- 3.Create a Table
mysql> CREATE TABLE employees (
-> id INT AUTO INCREMENT PRIMARY KEY,
-> first_name VARCHAR(50),
-> last_name VARCHAR(50),
-> department VARCHAR(50),
-> salary DECIMAL(10,2)
-> );
Query OK, 0 rows affected (0.05 sec)

mysql>
```

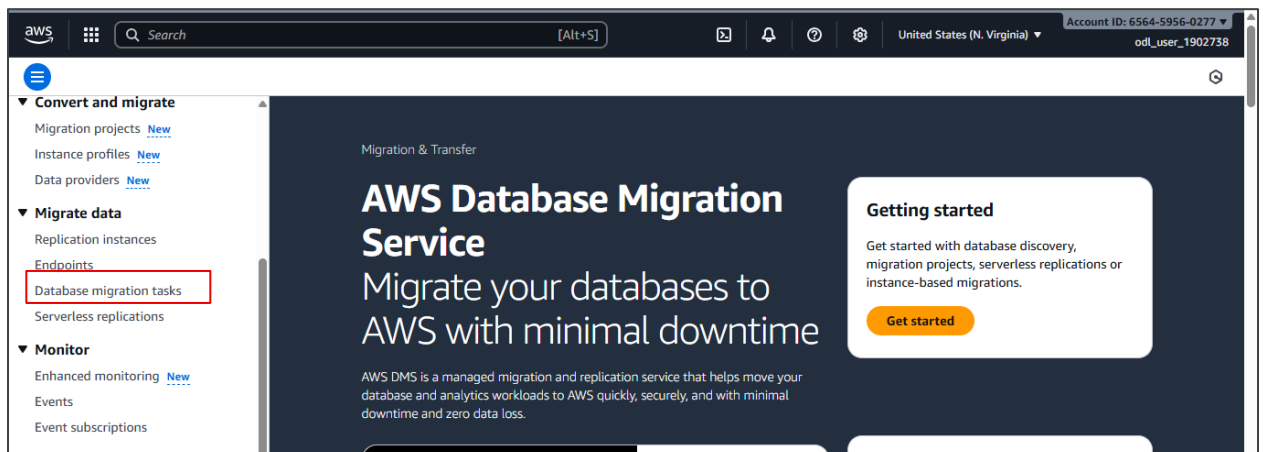
```
aws [Search] [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

mysql> INSERT INTO employees (first_name, last_name, department, salary)
-> VALUES
-> ('John', 'Doe', 'HR', 45000.00),
-> ('Jane', 'Smith', 'Finance', 55000.00),
-> ('Robert', 'Brown', 'IT', 65000.00),
-> ('Emily', 'Davis', 'Marketing', 50000.00),
-> ('Michael', 'Wilson', 'IT', 70000.00);
Query OK, 5 rows affected (0.01 sec)
Records: 5 Duplicates: 0 Warnings: 0

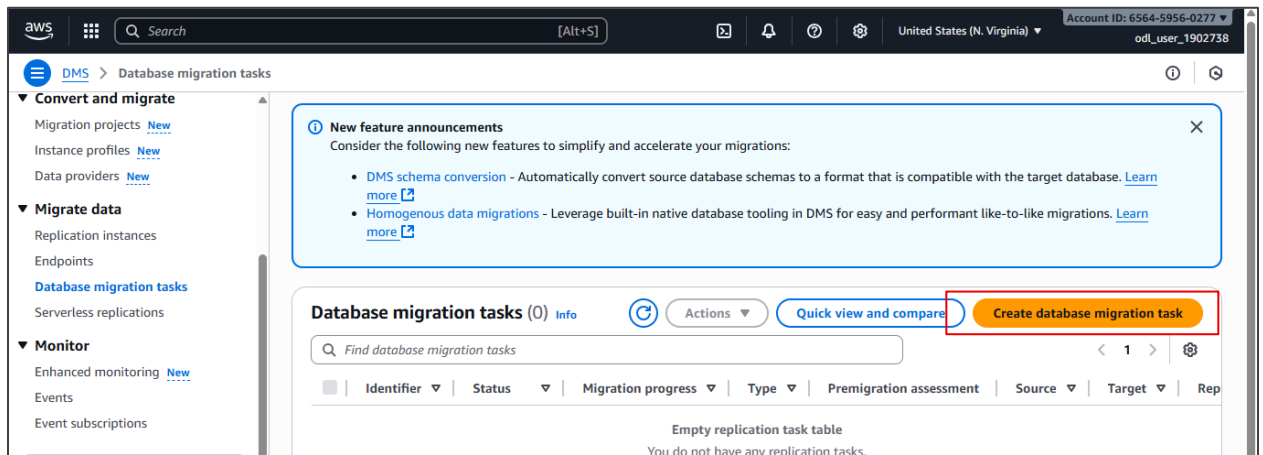
mysql> -- 5.Verify Data
mysql> SELECT * FROM employees;
+----+-----+-----+-----+-----+
| id | first_name | last_name | department | salary |
+----+-----+-----+-----+-----+
| 1  | John      | Doe      | HR         | 45000.00 |
| 2  | Jane      | Smith    | Finance    | 55000.00 |
| 3  | Robert    | Brown    | IT         | 65000.00 |
| 4  | Emily     | Davis    | Marketing  | 50000.00 |
| 5  | Michael   | Wilson   | IT         | 70000.00 |
+----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

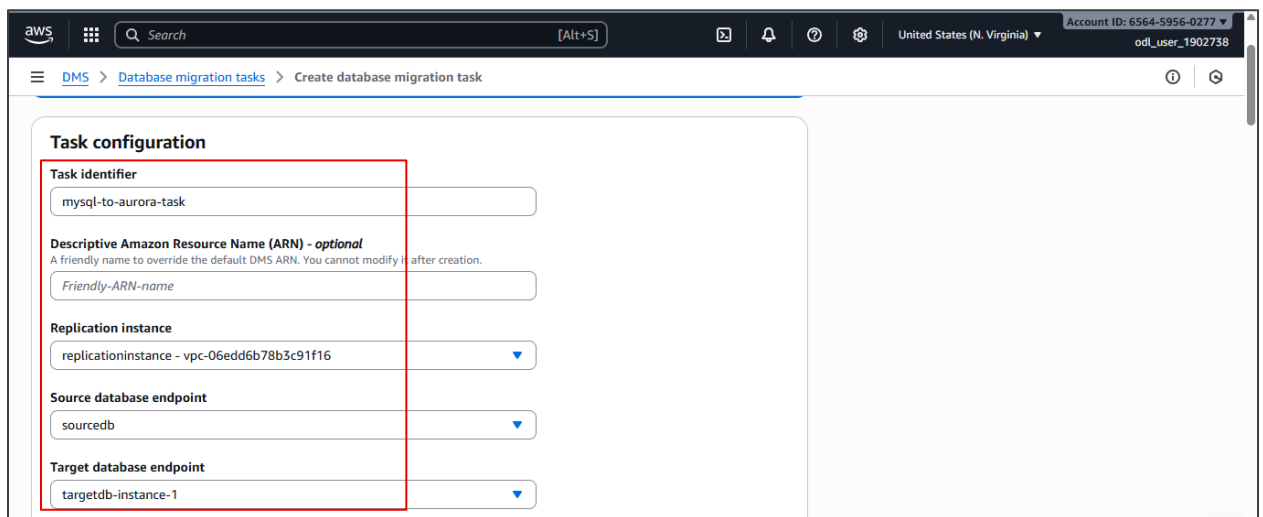
3.4 Click Database migration tasks



3.5 Click **Create database migration task**



3.6 Enter **mysql-to-aurora-task** for **Task identifier**, select **replicationinstance - vpc - 06edd6b78b3c91f16** for **Replication instance**, select **sourcedb** for **Source database endpoint**, and select **targetdb-instance-1** as **Target database endpoint**



3.7 Select **Migrate** for **Migration type** and enable **Wizard**

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Database migration tasks > Create database migration task

Migration type Info

- ☒ **Migrate**
Migrate data from source to target once
- ☐ **Migrate and replicate**
Migrate data from source to target once and continue to replicate changes
- ☐ **Replicate**
Migrate data from source to target from now, or at specified milestone changes

Task settings

Editing mode Info

- ☒ **Wizard**
You can enter only a subset of the available task settings.
- ☐ **JSON editor**
You can enter all available task settings directly in JSON format.

Target table preparation mode Info

- ☒ **Do nothing**
- ☐ Drop tables on target
- ☐ Truncate

3.8 Click **Add new selection rule**

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Database migration tasks > Create database migration task

Table mappings Info

Editing mode

- ☒ **Wizard**
You can enter only a subset of the available table mappings.
- ☐ **JSON editor**
You can enter all available table mappings directly in JSON format.

Specify at least one selection rule with an include action. After you do this, you can add one or more transformation rules.

▼ **Selection rules**

Choose the schema and/or tables you want to include with, or exclude from, your migration task.

Add new selection rule

3.9 Select **Enter a schema** for Schema

aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

DMS > Database migration tasks > Create database migration task

your migration task.

▼ where schema name is like '%' and Source table name is like '%', include

Schema

Enter a schema ▼

Source name
Use the % character as a wildcard

%

Source table name
Use the % character as a wildcard

%

3.10 Uncheck Turn on premigration assessment and enable Automatically on create

aws | Search [Alt+S] | United States (N. Virginia) | Account ID: 6564-5956-0277 | odl_user_1902738

DMS > Database migration tasks > Create database migration task

Feedback for premigration assessment
Can you share why you turned off the feature? This will help us improve your experience. [Provide feedback](#)

☐ **Turn on premigration assessment**
Choose this setting if you want AWS DMS to identify potential issues with the migration before starting the migration. Available only if replication instance, source, and target database are specified. After the current migration task is created, you can go to the Premigration assessments tab in the migration task details page to view assessment results or create a new assessment run with different settings.

► **Advanced settings**

Migration task startup configuration

Start migration task

☒ **Automatically on create**
Available only if the premigration assessment is not enabled.

☐ **Manually later**

3.11 Click Create database migration task

aws | Search [Alt+S] | United States (N. Virginia) | Account ID: 6564-5956-0277 | odl_user_1902738

DMS > Database migration tasks > Create database migration task

☐ **Turn on premigration assessment**
Choose this setting if you want AWS DMS to identify potential issues with the migration before starting the migration. Available only if replication instance, source, and target database are specified. After the current migration task is created, you can go to the Premigration assessments tab in the migration task details page to view assessment results or create a new assessment run with different settings.

► **Advanced settings**

Migration task startup configuration

Start migration task

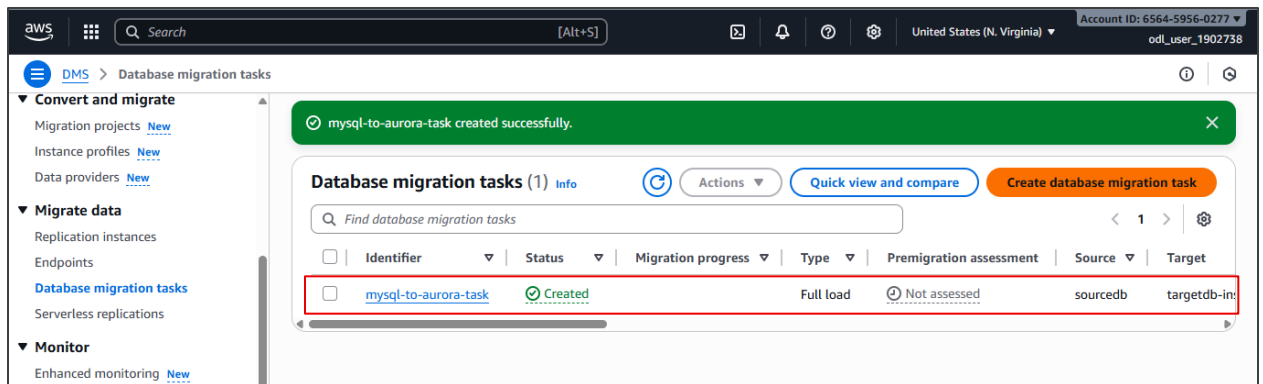
☒ **Automatically on create**
Available only if the premigration assessment is not enabled.

☐ **Manually later**

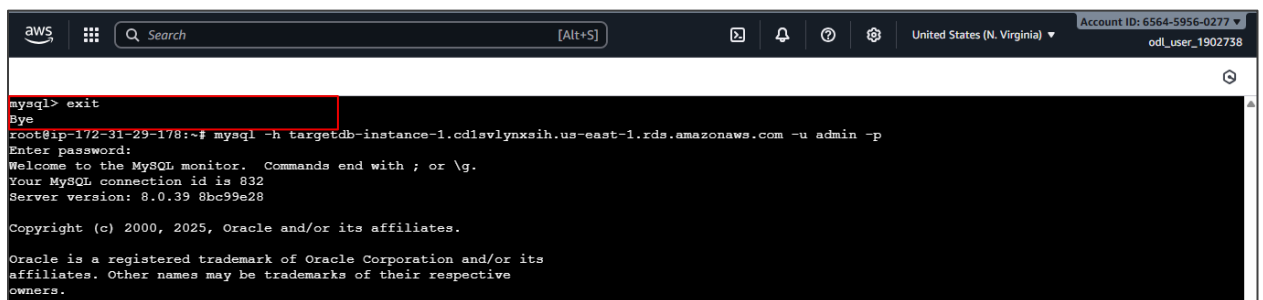
► **Tags**
Add tags to your DMS resources to organize and track your DMS costs.

[Cancel](#) [Create database migration task](#)

The following page appears:

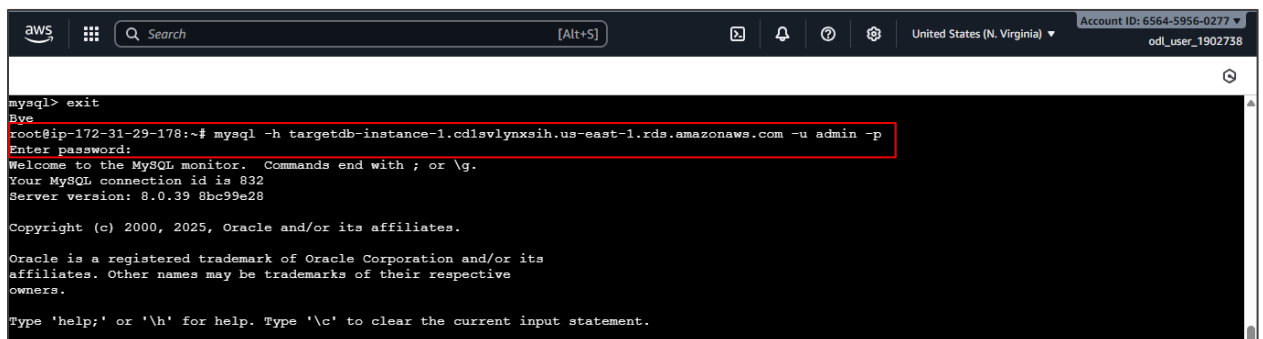


3.12 Navigate back to the terminal, type **exit**, and press the enter key



3.13 Run the following SQL command to connect targetdb, and press the enter key:

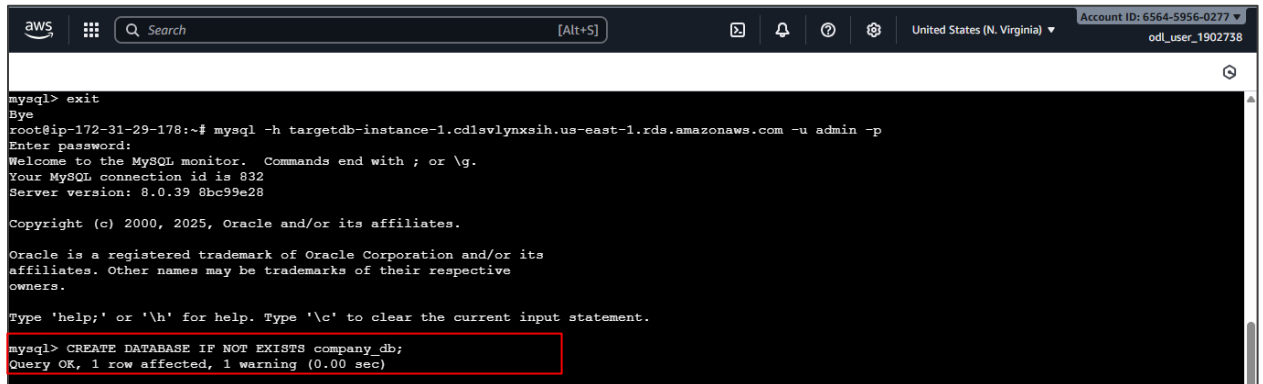
mysql -h targetdb-instance-1.cd1sylvjnxaih.us-east-1.rds.amazonaws.com -u admin -p



Note: The command will change based on the server name of sourcedb obtained in step 1.26. Format: **mysql -h <servername> -u admin -p**

3.14 Run the following SQL command to create the database if it does not exist:

CREATE DATABASE IF NOT EXISTS company_db;



```
mysql> exit
Bye
root@ip-172-31-29-178:~# mysql -h targetdb-instance-1.cdslvlynxsih.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 832
Server version: 8.0.39 8bc99e28

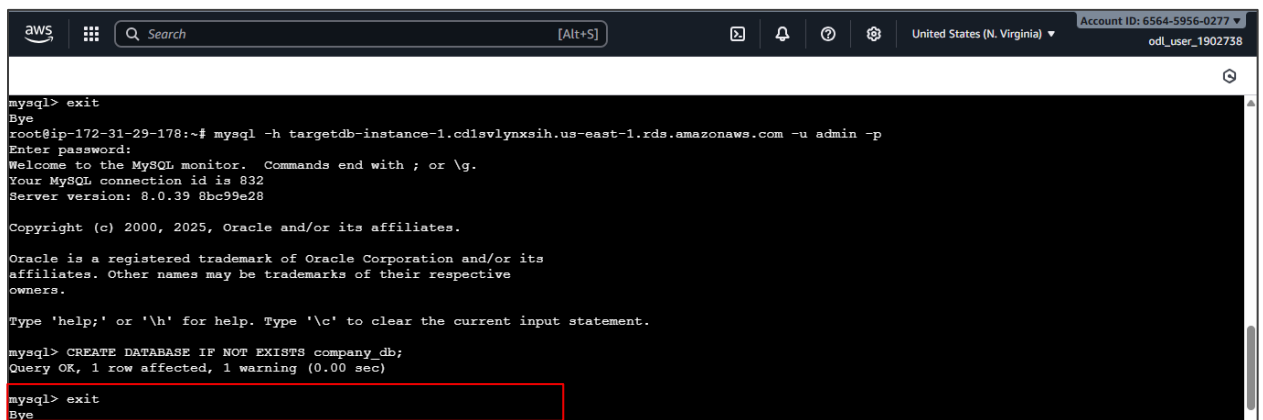
Copyright (c) 2000, 2025, Oracle and/or its affiliates.

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS company_db;
Query OK, 1 row affected, 1 warning (0.00 sec)
```

3.15 Type **exit**, and press the enter key



```
mysql> exit
Bye
root@ip-172-31-29-178:~# mysql -h targetdb-instance-1.cdslvlynxsih.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 832
Server version: 8.0.39 8bc99e28

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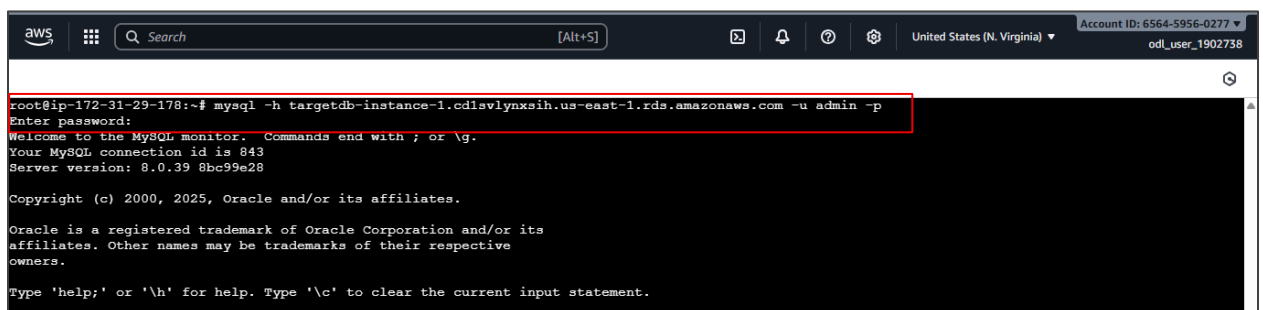
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE IF NOT EXISTS company_db;
Query OK, 1 row affected, 1 warning (0.00 sec)

mysql> exit
Bye
```

3.16 Run the following SQL command to connect targetdb, and enter the admin password:

mysql -h targetdb-instance1.cdslvlynxsih.us-east-1.rds.amazonaws.com -u admin -p



```
root@ip-172-31-29-178:~# mysql -h targetdb-instance1.cdslvlynxsih.us-east-1.rds.amazonaws.com -u admin -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 843
Server version: 8.0.39 8bc99e28

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
```

3.17 Run the following SQL command to use the database:

USE company_db;

```
aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
```

3.18 Run the following SQL command to print the list of all employees:
SELECT * FROM employees;

```
aws Search [Alt+S] United States (N. Virginia) Account ID: 6564-5956-0277 odl_user_1902738

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company_db;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM employees;
+----+-----+-----+-----+-----+
| id | first_name | last_name | department | salary |
+----+-----+-----+-----+-----+
| 1 | John      | Doe       | HR          | 45000.00 |
| 2 | Jane      | Smith     | Finance     | 55000.00 |
| 3 | Robert    | Brown     | IT           | 65000.00 |
| 4 | Emily     | Davis     | Marketing   | 50000.00 |
| 5 | Michael   | Wilson    | IT           | 70000.00 |
+----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

Note: In this step, we connect to the target RDS instance and create the database (company_db) that will serve as the destination. During the AWS DMS migration process, data from the source database is replicated into the target database, ensuring the schema and data are available for use in Aurora.

By following these steps, you successfully created the source and target database instances, set up a replication instance in AWS DMS, and executed the migration task to replicate data from the MySQL Community RDS instance to the AWS Aurora MySQL instance. This ensures that the target database (company_db) reflects the schema and records from the source, validating a complete and reliable database migration process.

